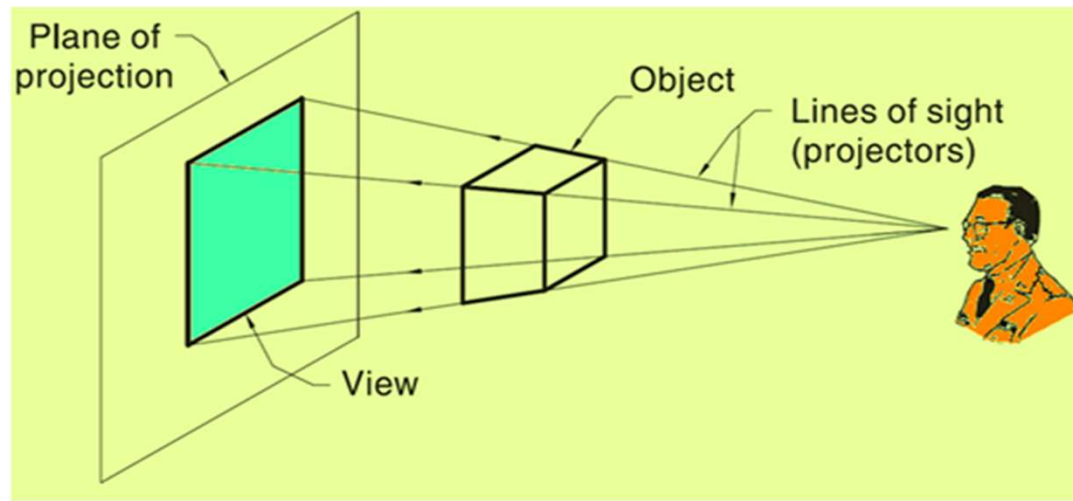


PROJECTION OF POINTS AND LINES

Theory of Projections

- In engineering, 3-dimensional objects and structures are represented graphically on a 2-dimensional media.
- The act of obtaining the image of an object is termed “projection”. The image obtained by projection is known as a “view”.
- In effect, 3-D object is transformed into a 2-D representation, also called projections. The paper or computer screen on which a drawing is created is a plane of projection.

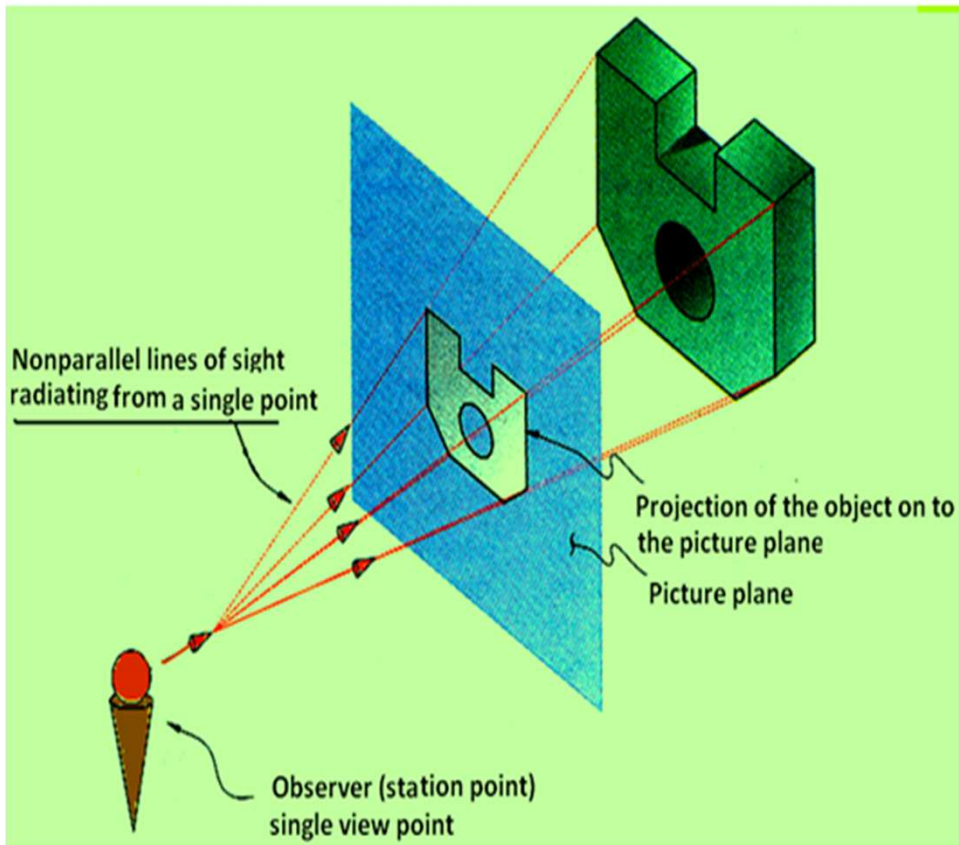


Theory of Projections (Contd...)

Plane of Projection

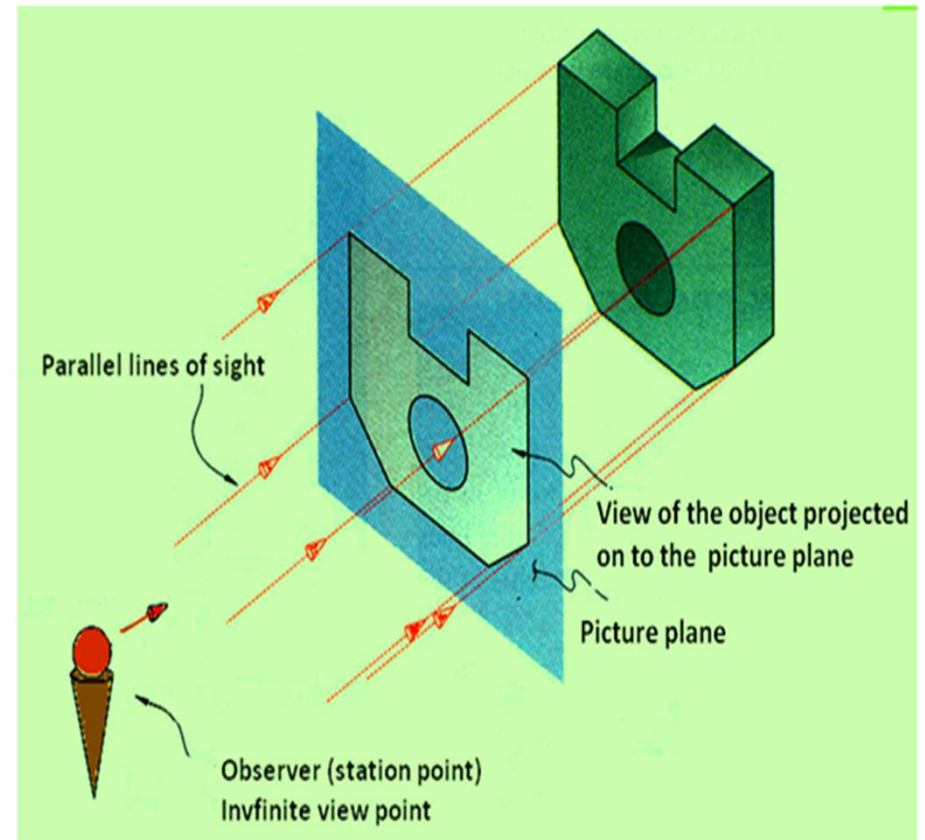
- A plane of projection (i.e, an image or picture plane) is an imaginary flat plane upon which the image created by the line of sight is projected.
- The image is produced by connecting the points where the lines of sight pierce the projection plane.
- The paper or computer screen on which a drawing is created is a plane of projection.

• Projection Methods



Perspective

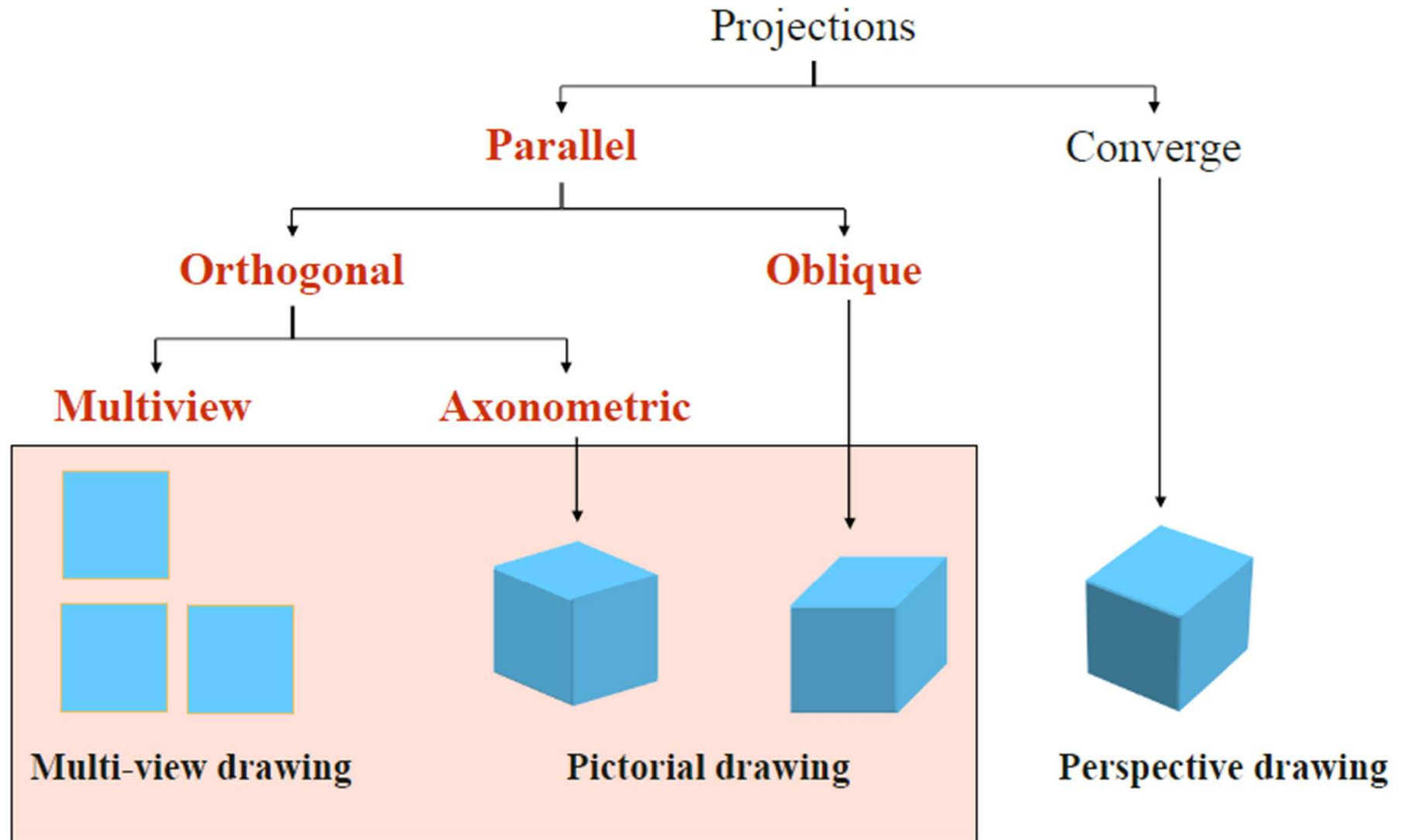
- Distance from the observer to the object is finite
- Projectors are not parallel
- Perspective projections mimic what the human eyes see, however, they are difficult to draw



Parallel

- Distance from the observer to the object is infinite
- object is positioned at infinity
- Less realistic but easier to draw

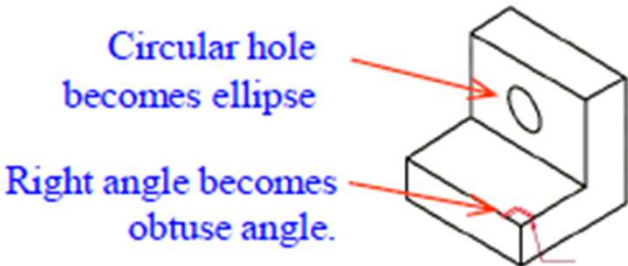
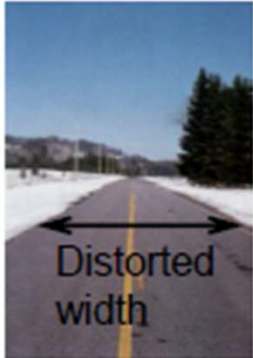
Type of Projections



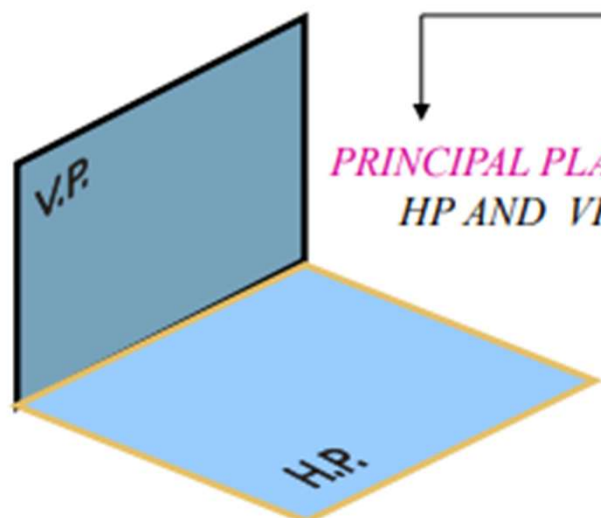
Orthographic Multi view Projections

- Orthographic multiview projections is a technical drawing in which different views of an object are projected on different reference planes observing perpendicular to respective reference plane.
- Different Reference planes are;
 - Horizontal Plane (HP)
 - Vertical Plane (VP)
 - Side or Profile Plane (PP)
- Different views are;
 - *Front View (FV) –Projected on VP*
 - *Top View (TV) –Projected on HP*
 - *Side View (SV) –Projected on PP*

View comparison

Type		
Multi-view drawing 	● Accurately presents object's details, i.e. size and shape.	● Require training to visualization.
Pictorial drawing 	● Easy to visualize.	● Shape and angle distortion Circular hole becomes ellipse Right angle becomes obtuse angle. 
Perspective drawing 	● Object looks more like what our eyes perceive.	● Difficult to create ● Size and shape distortion 

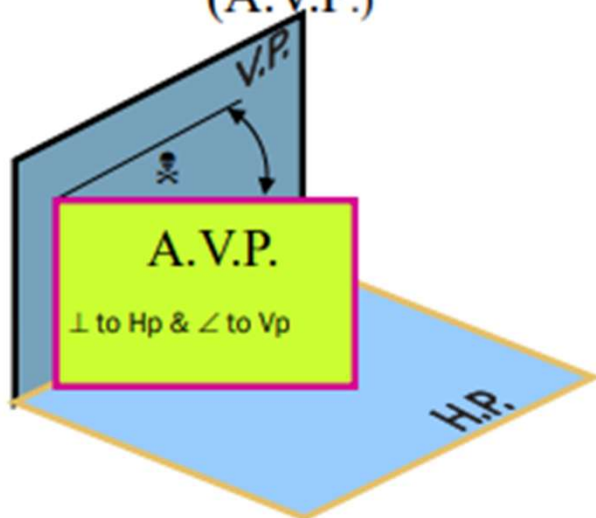
PLANES



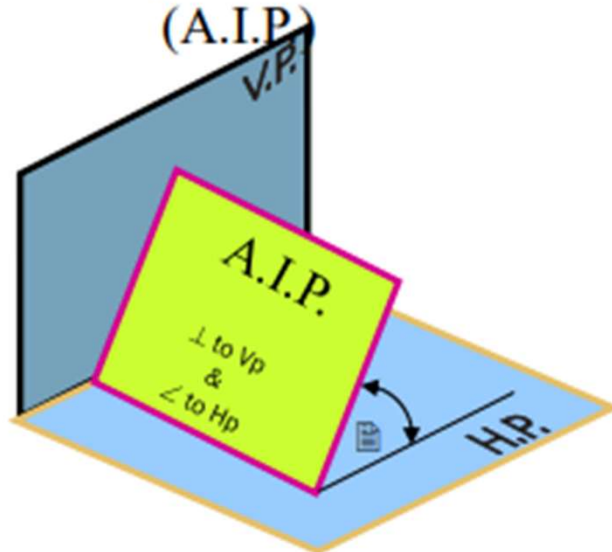
PRINCIPAL PLANES
HP AND VP

AUXILIARY PLANES

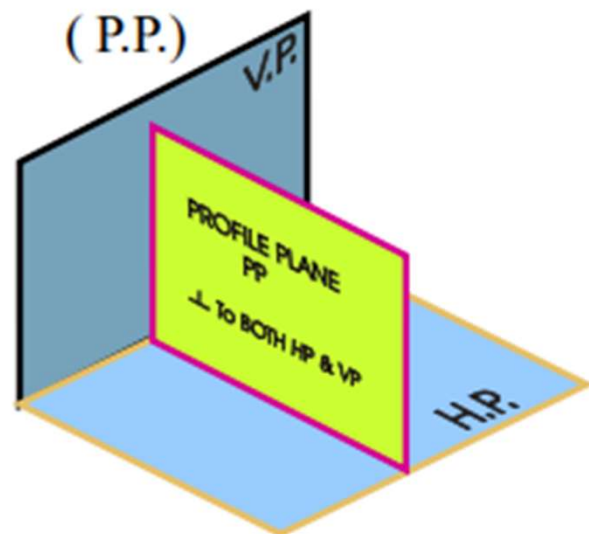
Auxiliary Vertical Plane
(A.V.P.)



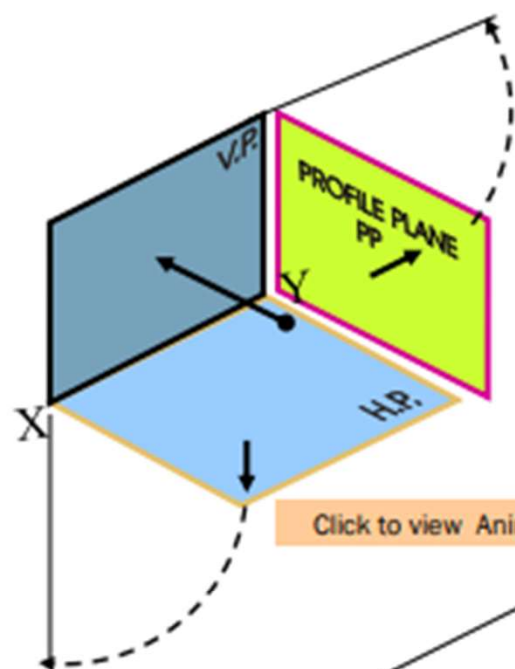
Auxiliary Inclined Plane
(A.I.P.)



Profile Plane
(P.P.)



PATTERN OF PLANES & VIEWS (First Angle Method)



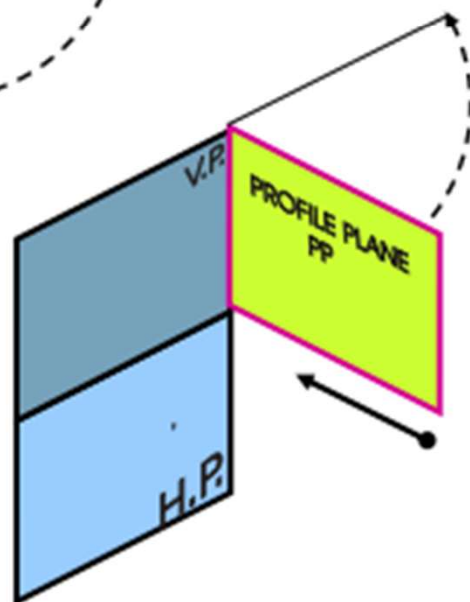
THIS IS A PICTORIAL SET-UP OF ALL THREE PLANES.
ARROW DIRECTION IS A NORMAL WAY OF OBSERVING THE OBJECT.
BUT IN THIS DIRECTION ONLY VP AND A VIEW ON IT (FV) CAN BE SEEN.
THE OTHER PLANES AND VIEWS ON THOSE CAN NOT BE SEEN.

PROCEDURE TO SOLVE ABOVE PROBLEM:-

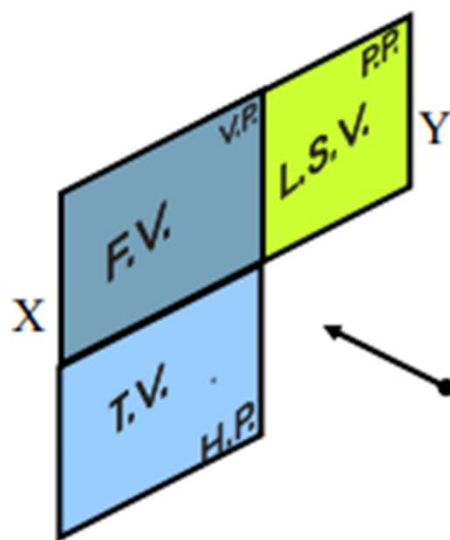
TO MAKE THOSE PLANES ALSO VISIBLE FROM THE ARROW DIRECTION,
A) HP IS ROTATED 90° DOWNWARD
B) PP, 90° IN RIGHT SIDE DIRECTION.
THIS WAY BOTH PLANES ARE BROUGHT IN THE SAME PLANE CONTAINING VP.

Click to view Animation

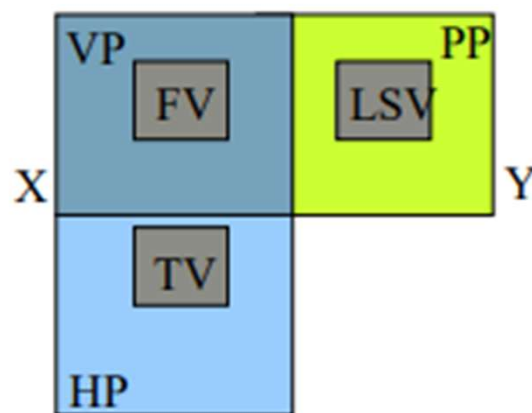
On clicking the button if a warning comes please click YES to continue, this program is safe for your pc.



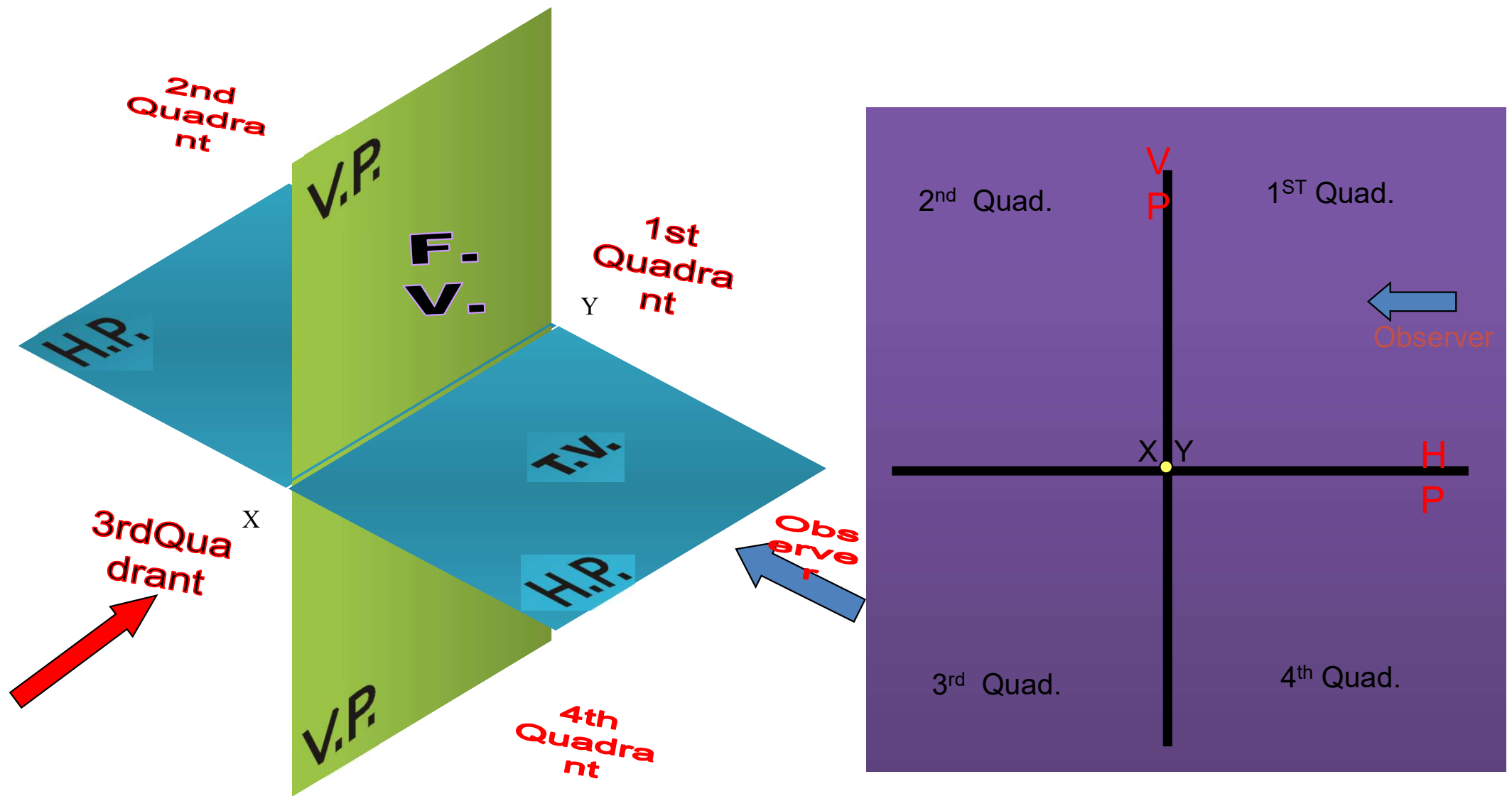
HP IS ROTATED DOWNWARD 90°
AND
BROUGHT IN THE PLANE OF VP.



PP IS ROTATED IN RIGHT SIDE 90°
AND
BROUGHT IN THE PLANE OF VP.



ACTUAL PATTERN OF PLANES & VIEWS
OF ORTHOGRAPHIC PROJECTIONS
DRAWN IN
FIRST ANGLE METHOD OF PROJECTIONS



THIS QUADRANT PATTERN,
IF OBSERVED ALONG X-Y LINE (IN RED ARROW DIRECTION)
WILL EXACTLY APPEAR AS SHOWN ON RIGHT SIDE AND HENCE,
IT IS FURTHER USED TO UNDERSTAND ILLUSTRATION PROPERLLY.

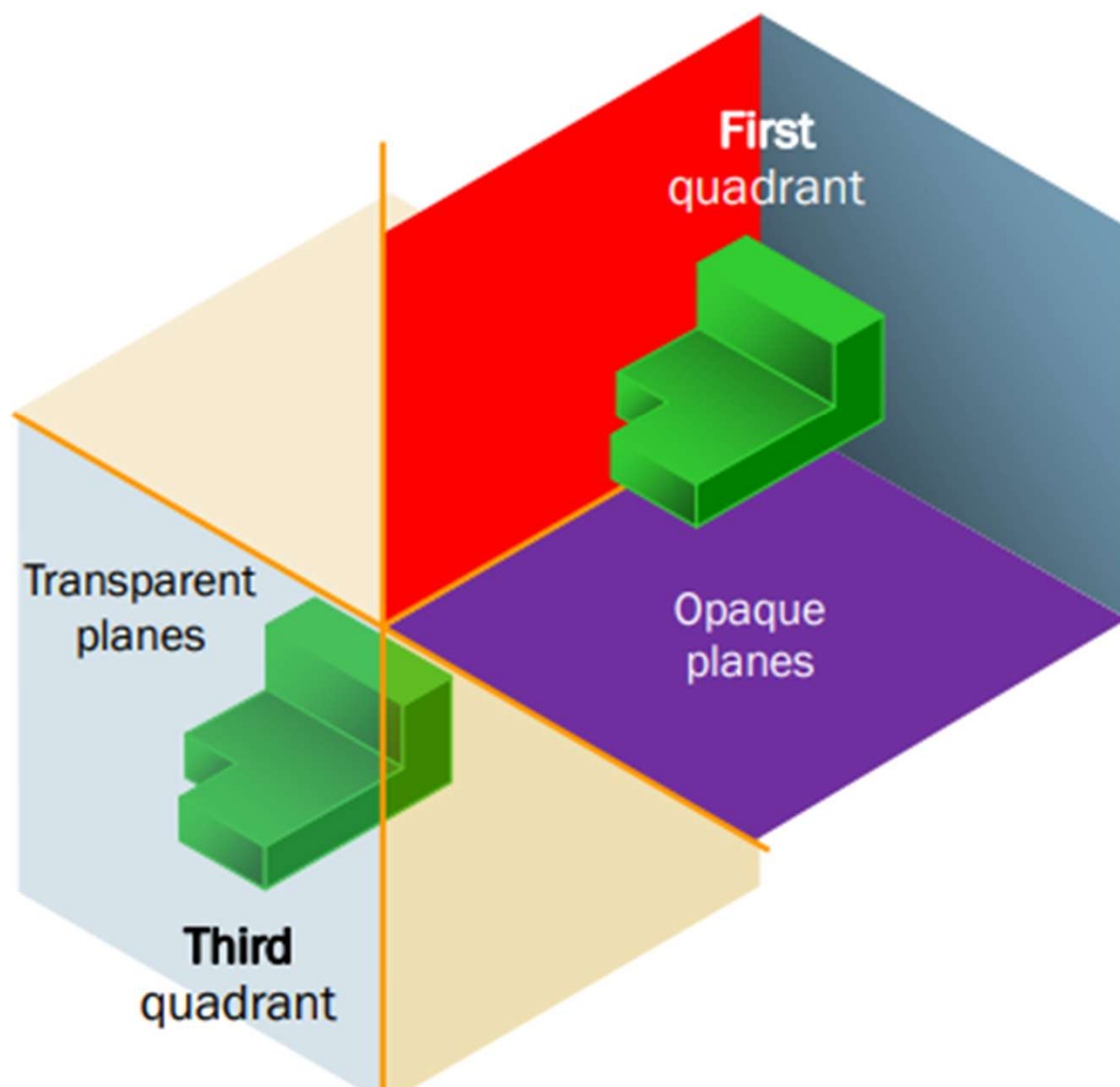
Projection systems

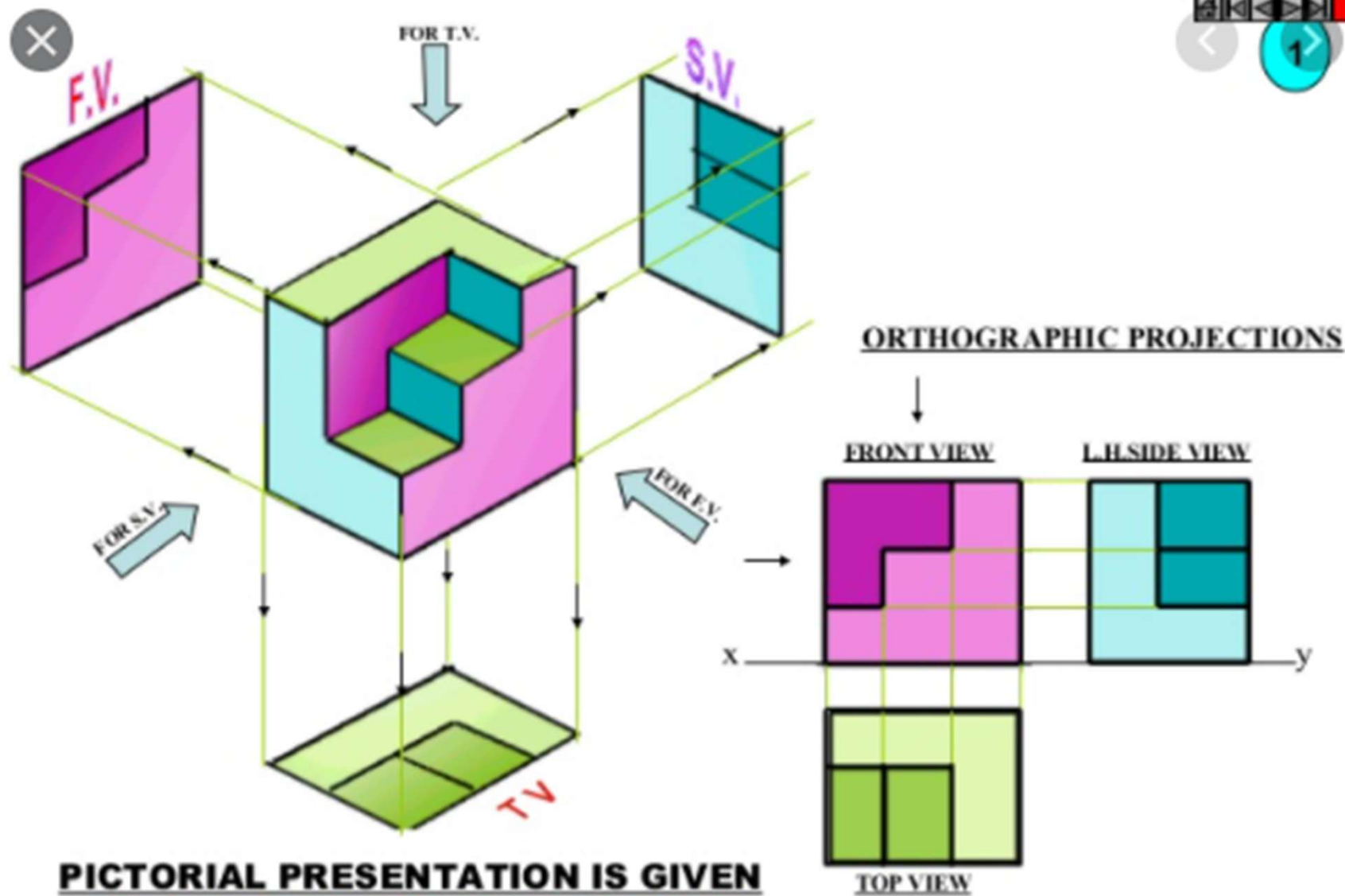
1. **First** angle system

- European countries
- ISO standard

2. **Third** angle system

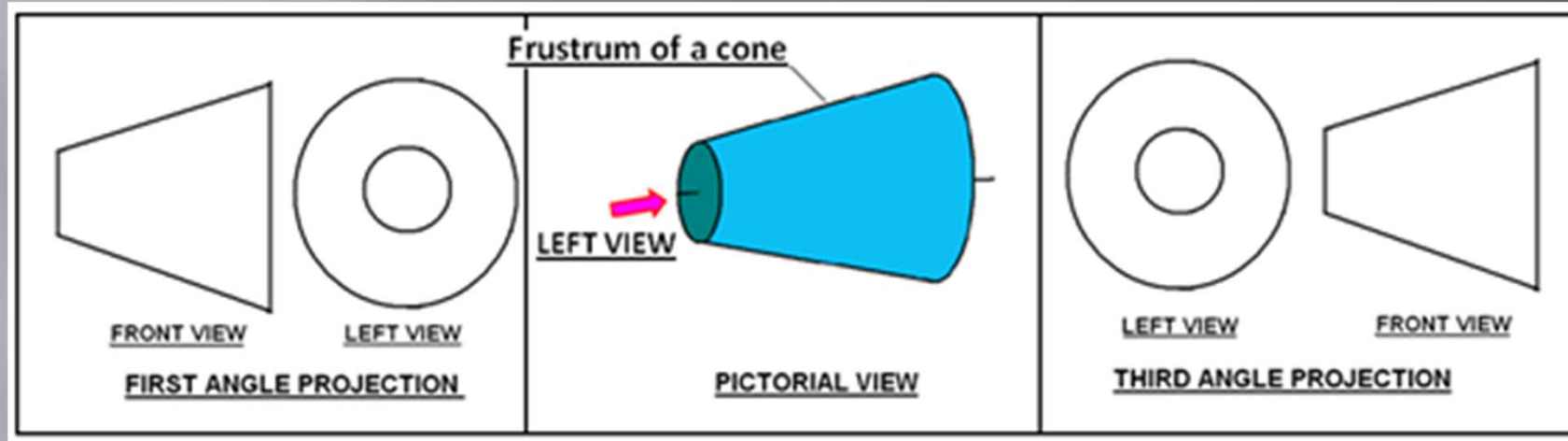
- Canada, USA,
Japan, Thailand





PICTORIAL PRESENTATION IS GIVEN
DRAW THREE VIEWS OF THIS OBJECT
BY FIRST ANGLE PROJECTION METHOD

Symbol of projection



NOTATIONS

FOLLOWING NOTATIONS SHOULD BE FOLLOWED WHILE NAMING DIFFERENT VIEWS IN ORTHOGRAPHIC PROJECTIONS.

OBJECT	POINT A	LINE AB
IT'S TOP VIEW	a	a b
IT'S FRONT VIEW	a'	a' b'
IT'S SIDE VIEW	a''	a'' b''

***SAME SYSTEM OF NOTATIONS SHOULD BE FOLLOWED
INCASE NUMBERS, LIKE 1, 2, 3 – ARE USED.***

Orientation of Point in Space

- (1) In quadrant *I* (Above H.P & In Front of V.P.)**
- (2) In quadrant *II* (Above H.P & Behind V.P.)**
- (3) In quadrant *III* (Below H.P & Behind V.P.)**
- (4) In quadrant *IV* (Below H.P & In Front of V.P.)**

Orientation of Point in Space

(5) In Plane (*Above H.P. & In V.P.*)

(6) In Plane (*Below H.P. & In V.P.*)

(7) In Plane (*In H.P. & In front of V.P.*)

(8) In Plane (*In H.P. & Behind V.P.*)

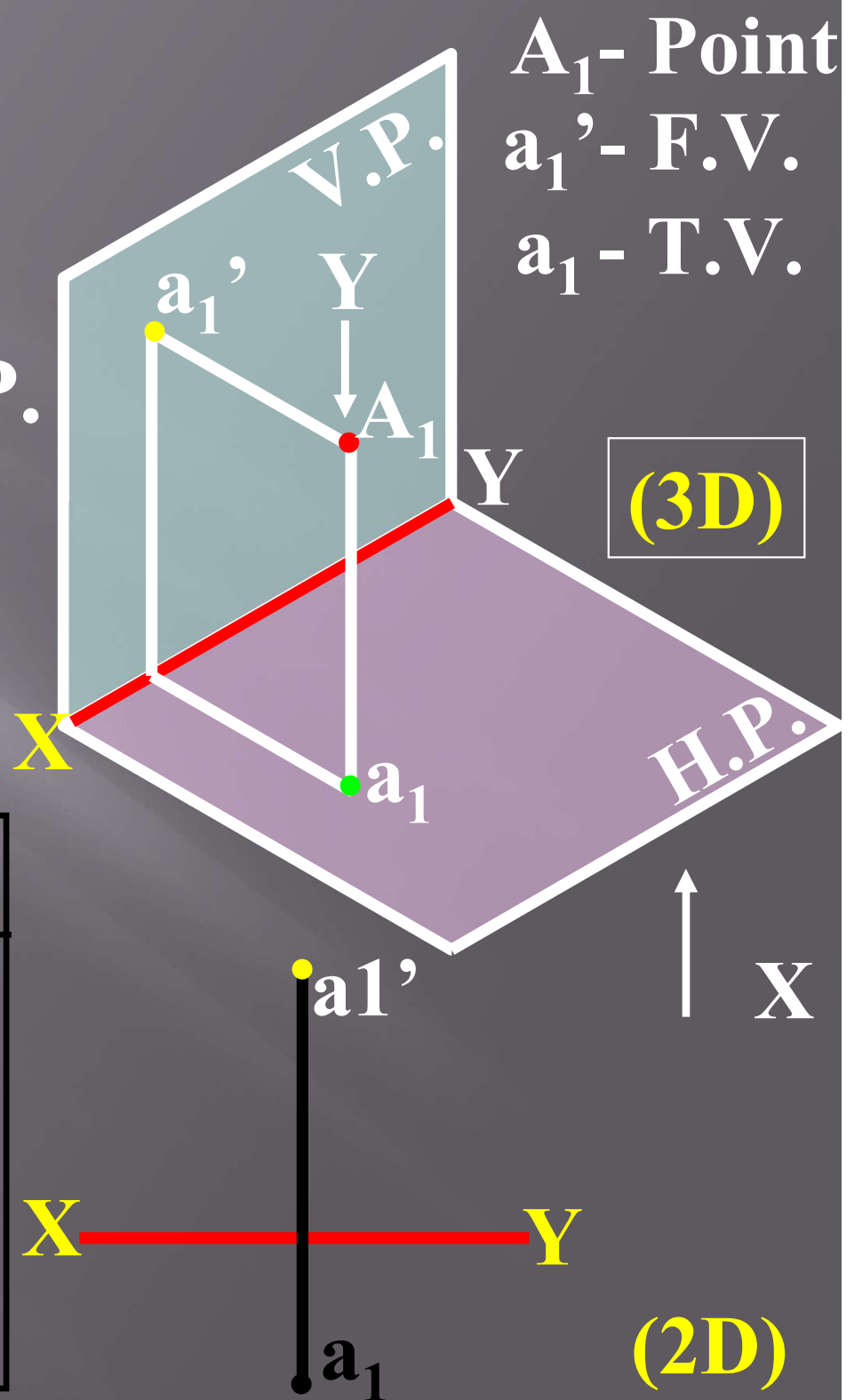
(9) In Plane (*In H.P. & V.P.*)

POSITION: 1 (I Qua.)

POINT A_1 $\left\{ \begin{array}{l} \text{Above H.P.} \\ \text{In Front Of V.P.} \end{array} \right.$

CONCLUSIONS:

<i>In 3D</i>	<i>In 2D</i>
Point, Above H.P.	F.V. Above XY
Point, In-Front Of V.P.	T.V. Below XY



POSITION:2 (II Qua.)

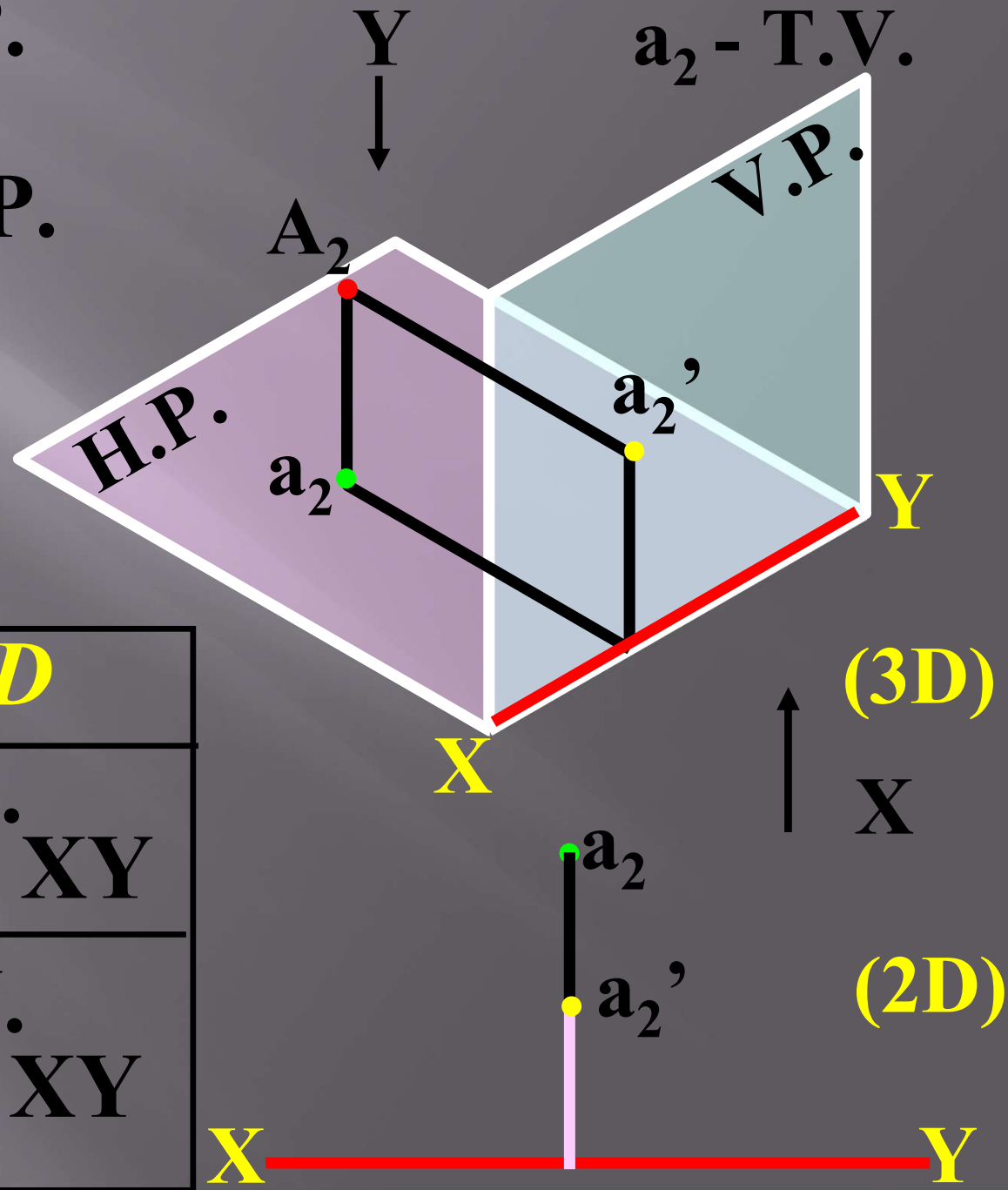
POINT A_2

- Above H.P.
- Behind V.P.

A_2 - Point
 a_2' - F.V.
 a_2 - T.V.

CONCLUSIONS:

In 3D	In 2D
Point, Above H.P.	F.V. Above XY
Point, Behind V.P.	T.V. Above XY

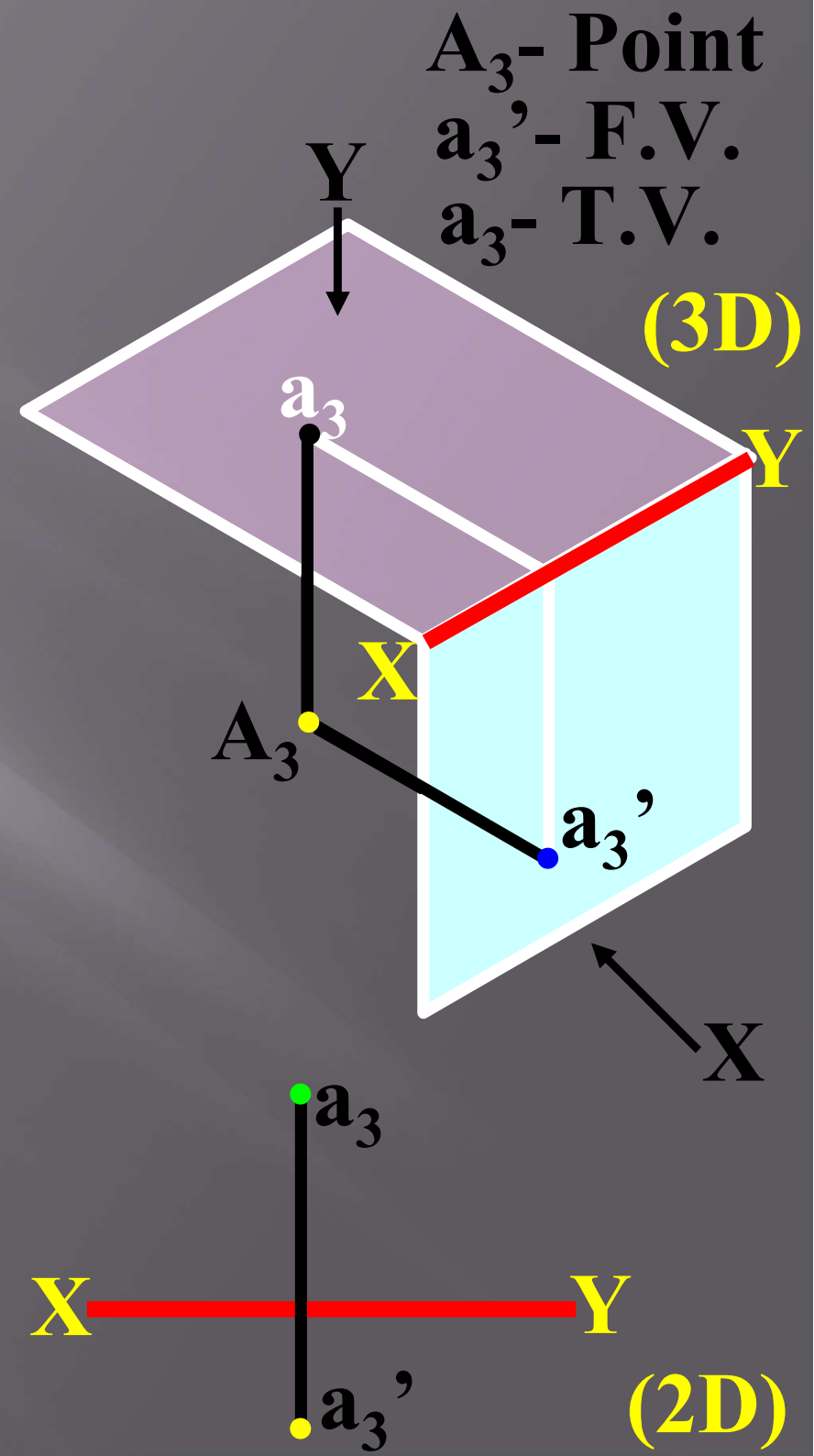


POSITION: 3 (III Qua.)

POINT
 A_3 { **Below H.P.**
{ **Behind V.P.**

CONCLUSIONS:

<i>In 3D</i>	<i>In 2D</i>
Point, Below H.P.	F.V. Below XY
Point Behind V.P.	T.V. Above XY

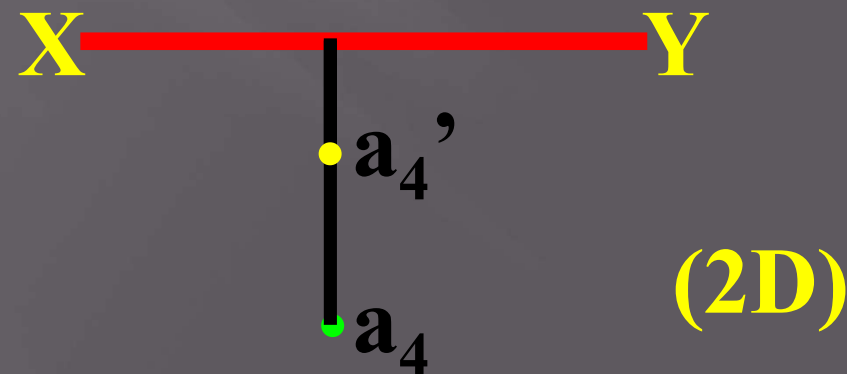
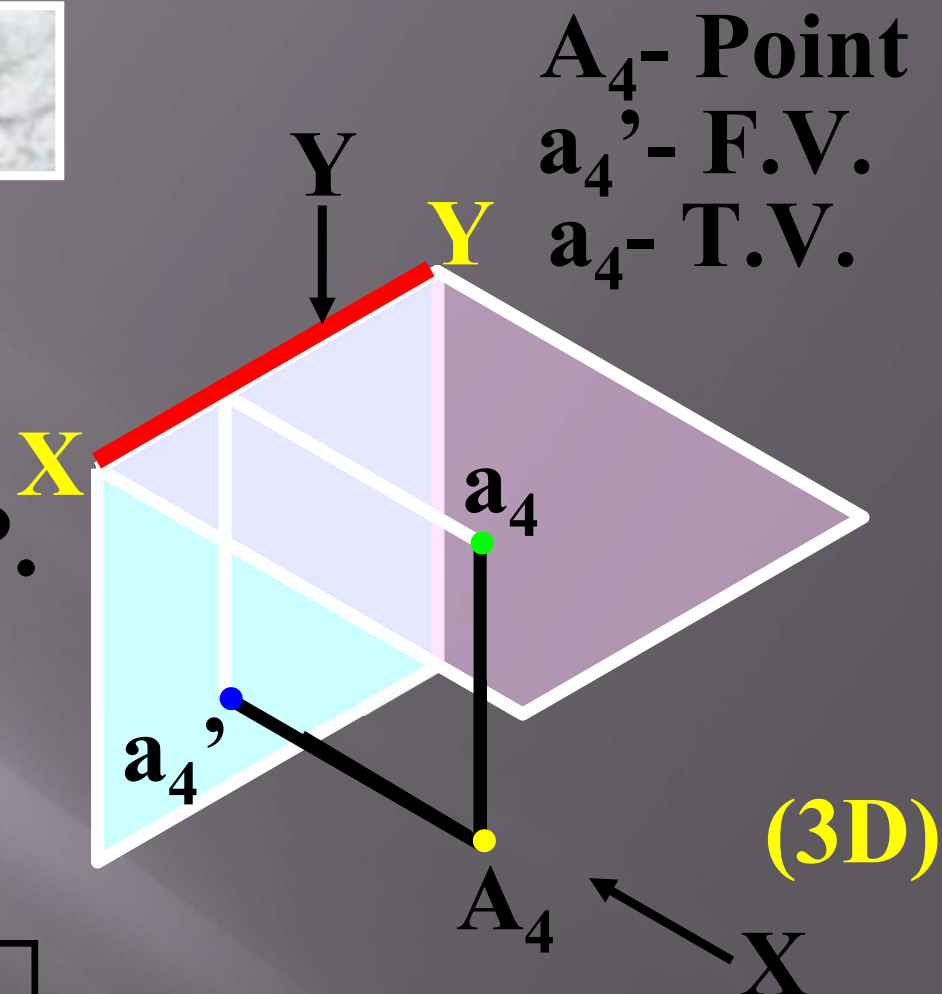


POSITION: 4 (IV Qua.)

POINT
 A_4 **Below H.P.**
In Front of V.P.

CONCLUSIONS:

<i>In 3D</i>	<i>In 2D</i>
Point, Below H.P.	F.V. Below XY
Point, In Front Of V.P.	T.V. Below XY

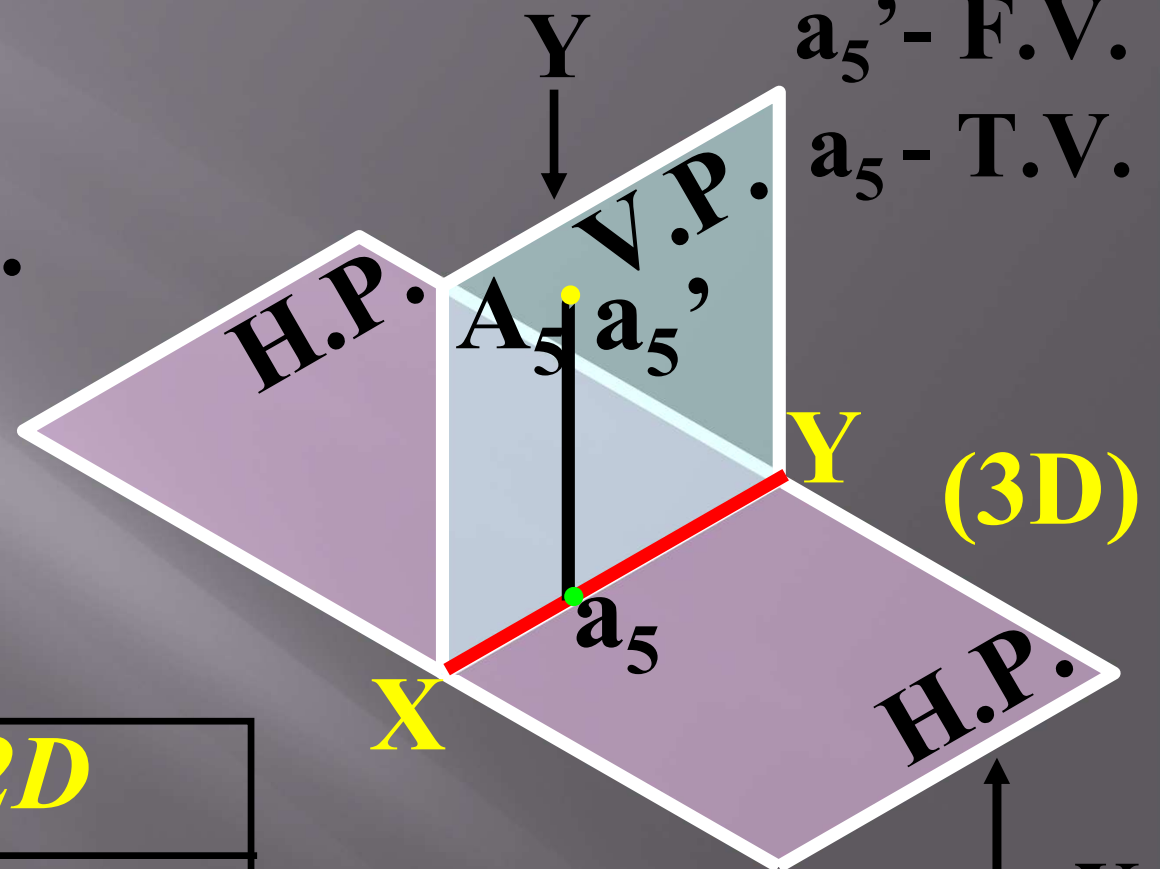


POSITION: 5

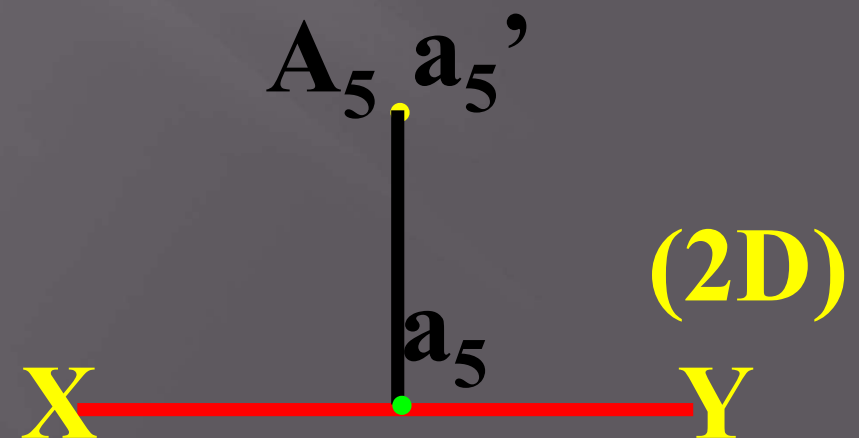
POINT A_5
 { Above H.P.
 { In V.P.

CONCLUSIONS:

In 3D	In 2D
Point, Above H.P.	F.V. Above XY
Point, In V.P.	T.V. On XY



A_5 - Point
 a_5' - F.V.
 a_5 - T.V.

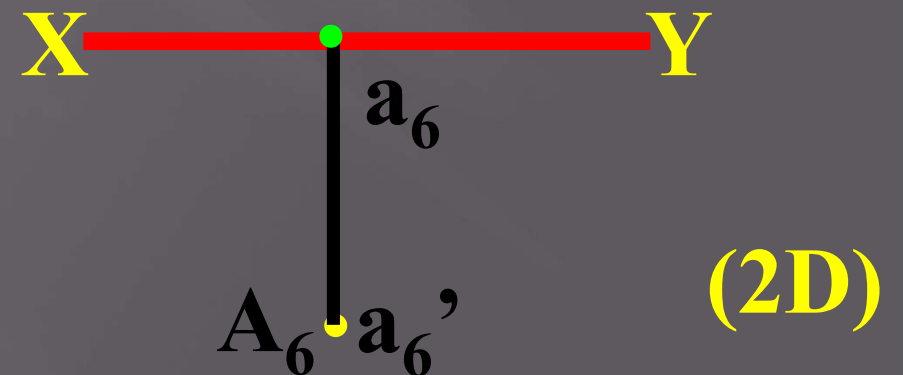
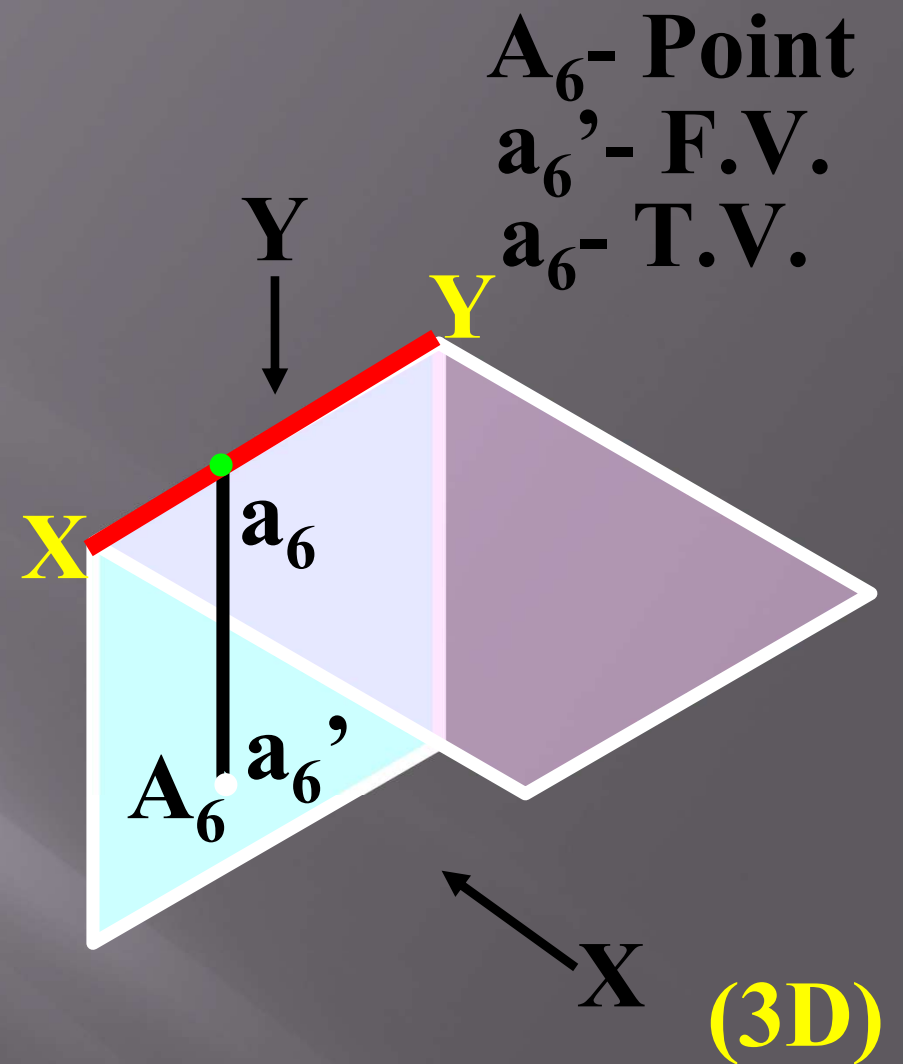


POSITION: 6

POINT
A₆

Below H.P.
In V.P.

<i>In 3D</i>	<i>In 2D</i>
Point, Below H.P.	F.V. Below XY
Point In V.P.	T.V. On XY

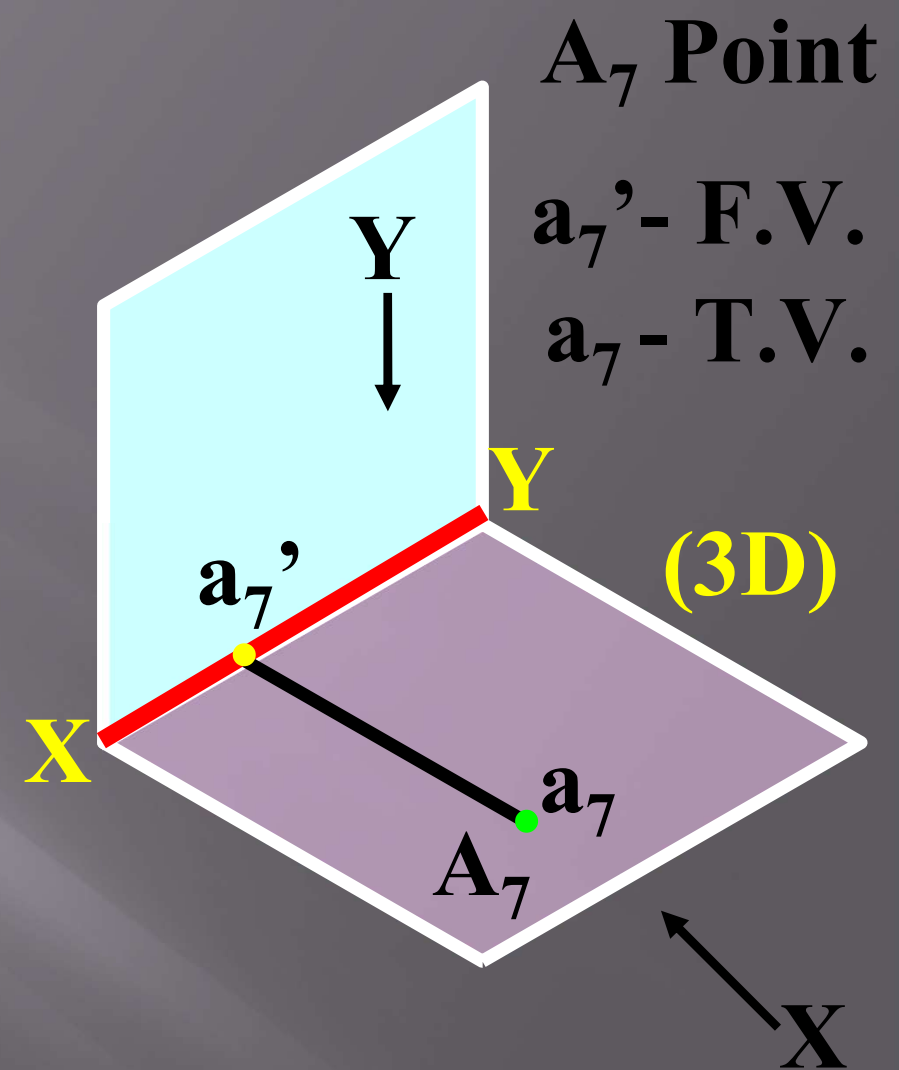


POSITION: 7

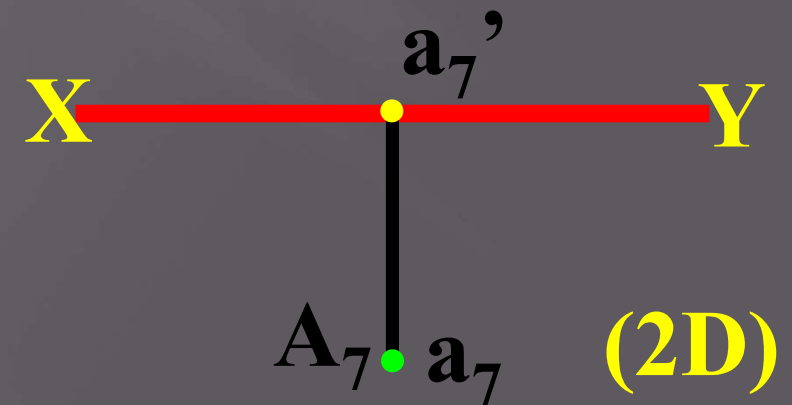
POINT A_7 $\left\{ \begin{array}{l} \text{In H.P.} \\ \text{In Front of V.P.} \end{array} \right.$

CONCLUSIONS:

In 3D	In 2D
Point, In-Front Of V.P.	T.V. Below XY
Point In H.P.	F.V. On XY



A_7 Point
 a_7' - F.V.
 a_7 - T.V.

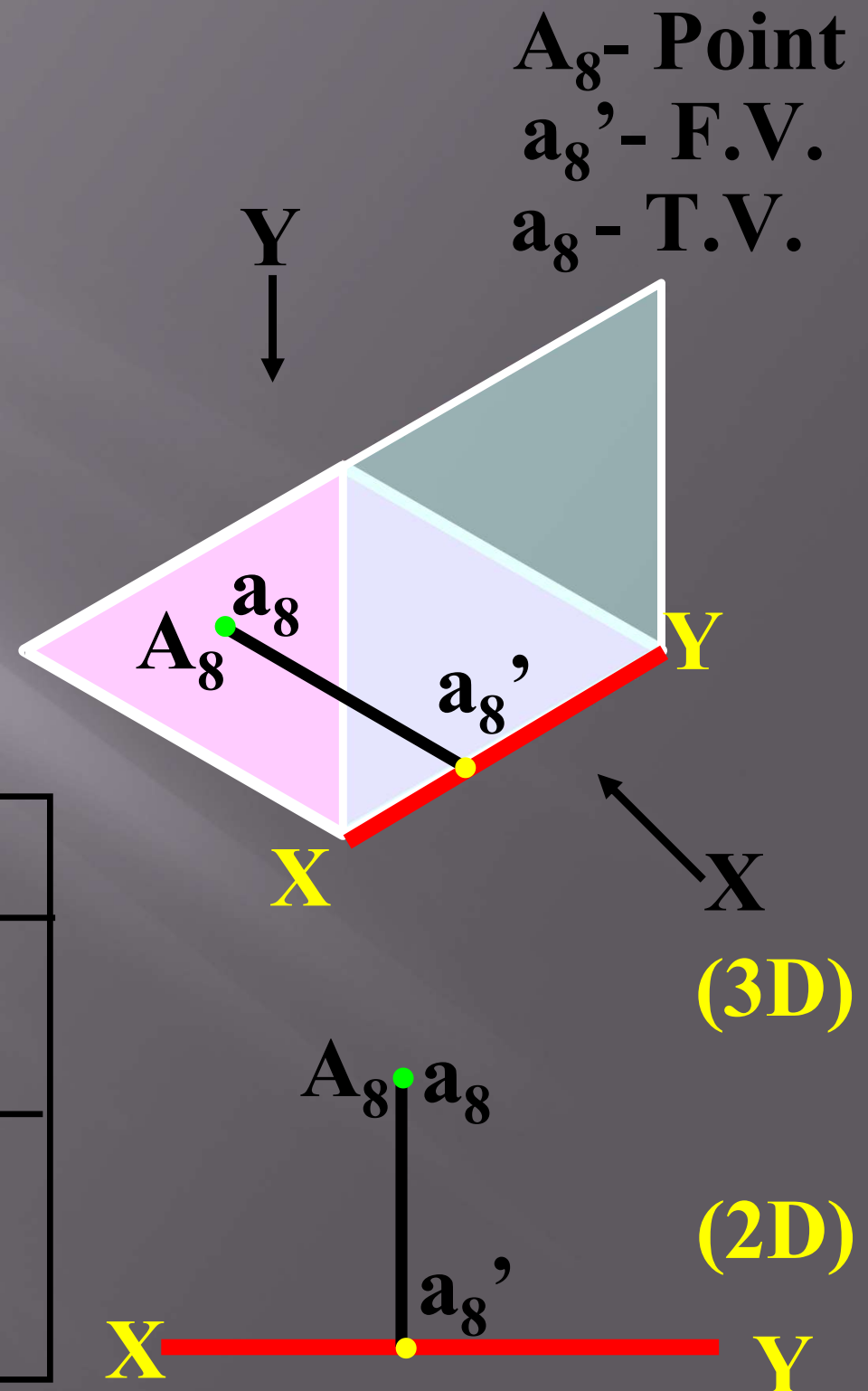


POSITION: 8

POINT A_8 { In H.P.
Behind V.P.

CONCLUSIONS:

<i>In 3D</i>	<i>In 2D</i>
Point, Behind V.P.	T.V. Above XY
Point, In H.P.	F.V. On XY

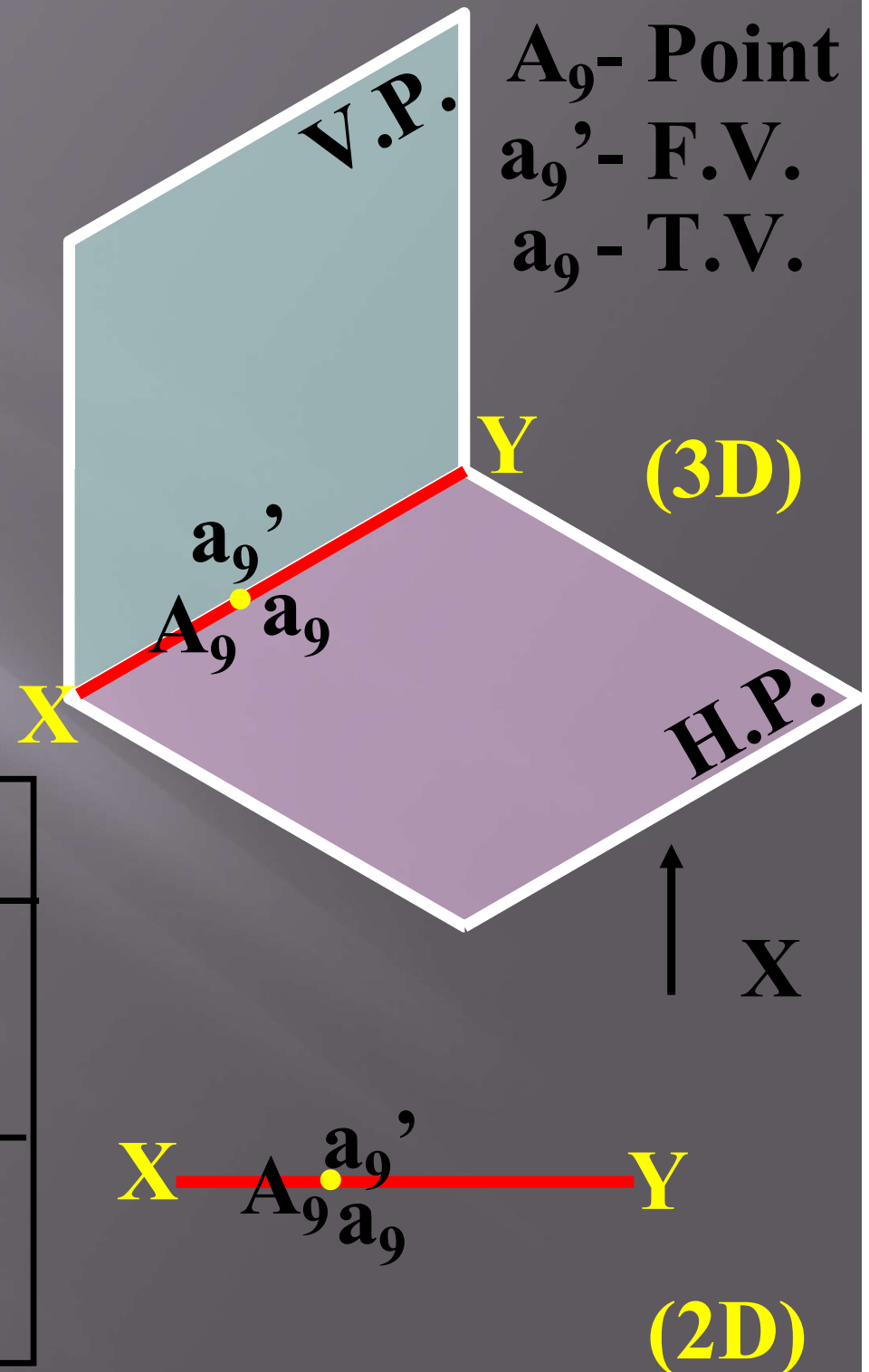


POSITION: 9

POINT A_9
 In H.P.
 In V.P.

CONCLUSIONS:

In 3D	In 2D
Point, In H.P.	F.V. On XY
Point, In V.P.	T.V. On XY



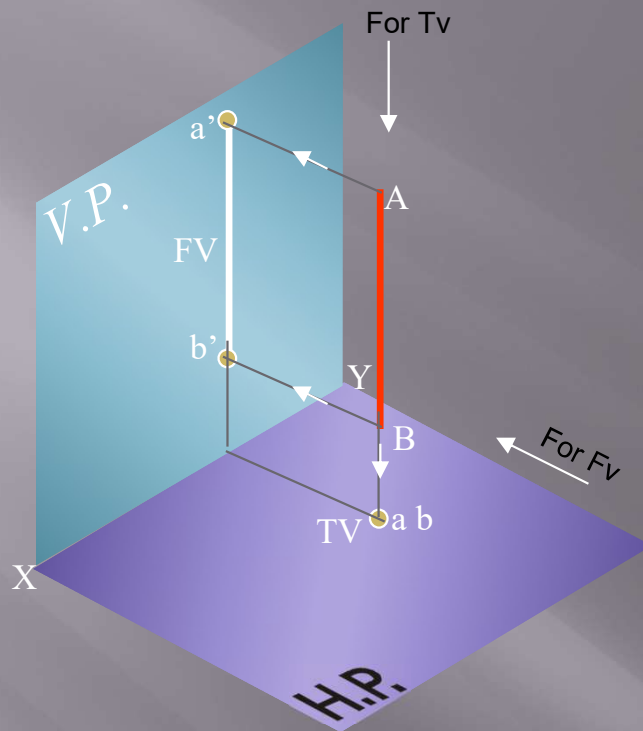
PROJECTIONS OF STRAIGHT LINES.

INFORMATION REGARDING A LINE *means*
IT'S LENGTH,
POSITION OF IT'S ENDS WITH HP & VP
IT'S INCLINATIONS WITH HP & VP WILL BE GIVEN.
AIM:- TO DRAW IT'S PROJECTIONS - MEANS FV & TV.

SIMPLE CASES OF THE LINE

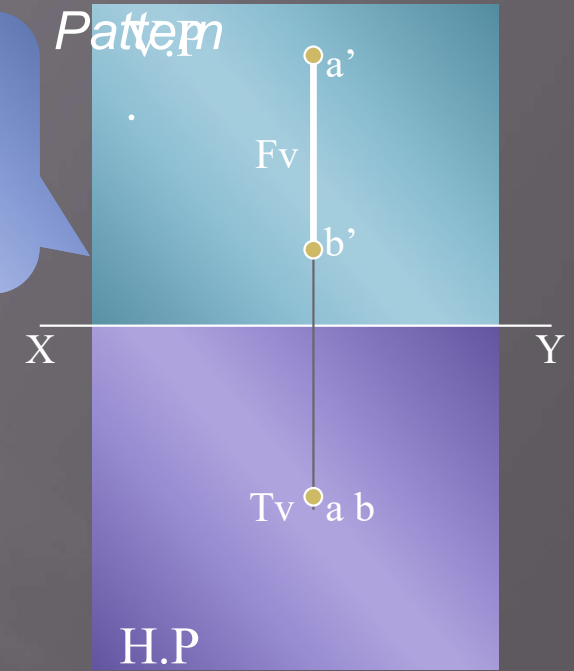
1. A VERTICAL LINE (LINE PERPENDICULAR TO HP & // TO VP)
1. LINE PARALLEL TO BOTH HP & VP.
1. LINE INCLINED TO HP & PARALLEL TO VP.
1. LINE INCLINED TO VP & PARALLEL TO HP.

1
A Line
perpendicular
to Hp
&
// to Vp

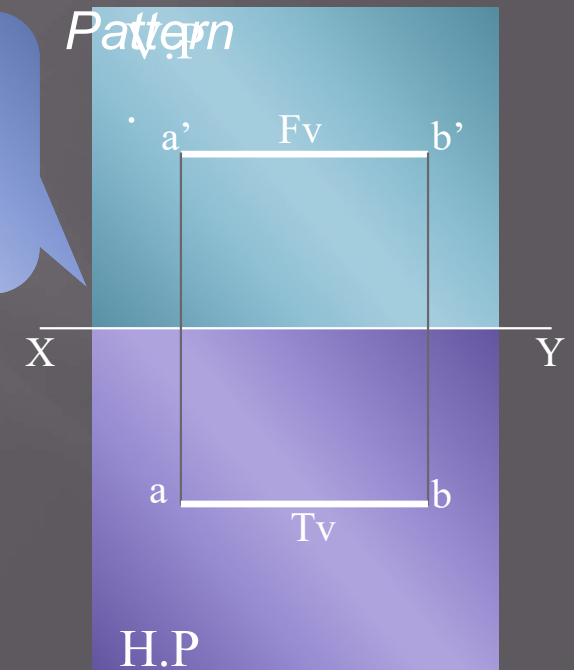


Fv is a vertical
line
Showing True
Length &
Tv is a point.

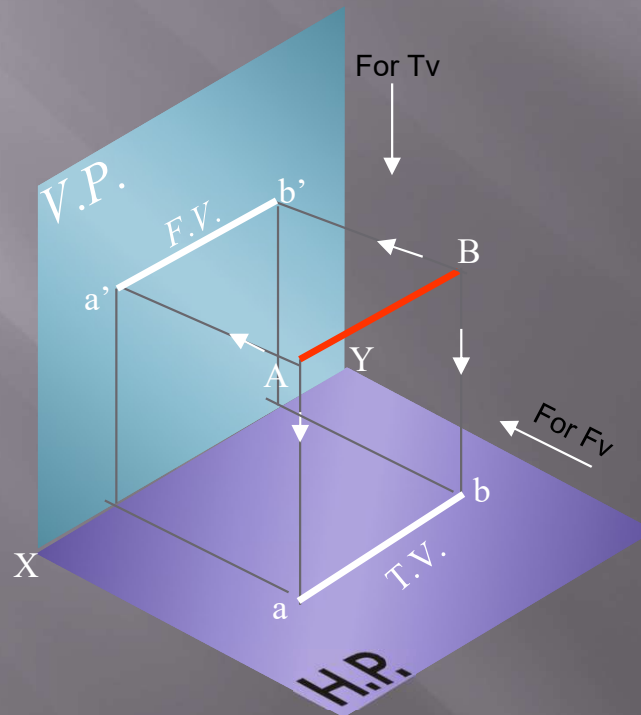
Orthographic
Pattern



Orthographic
Pattern



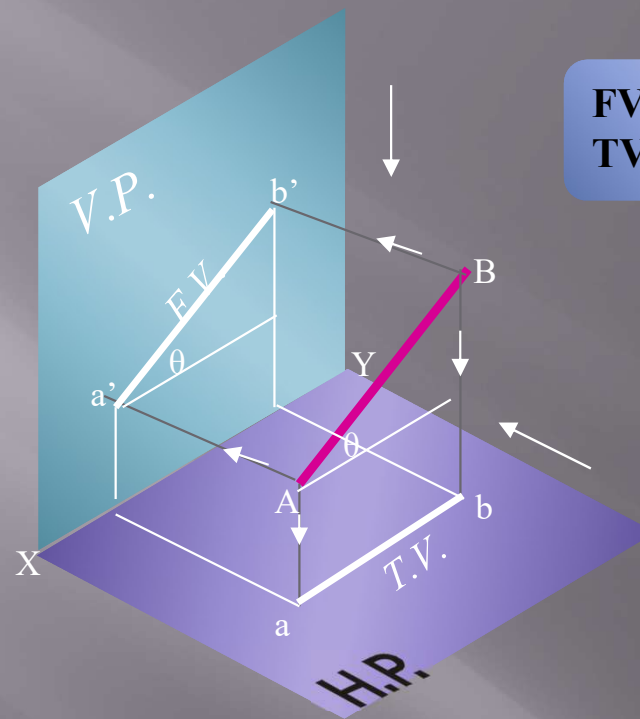
2
A Line
// to Hp
&
// to Vp



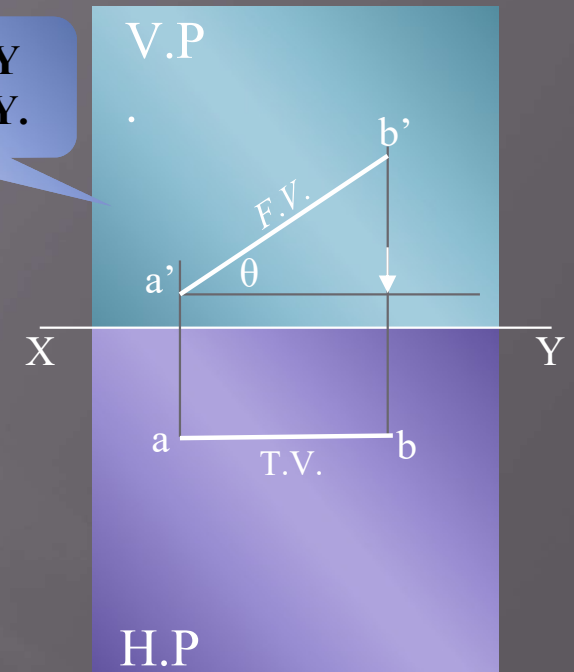
Fv & Tv both are
// to xy
&
both show T. L.

3

**A Line inclined to Hp
and
parallel to Vp**



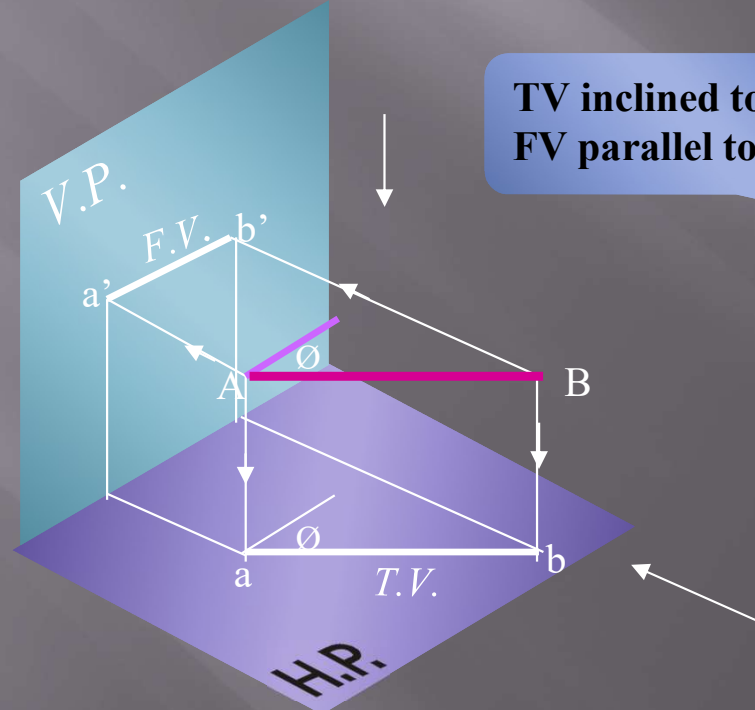
**FV inclined to XY
TV parallel to XY.**



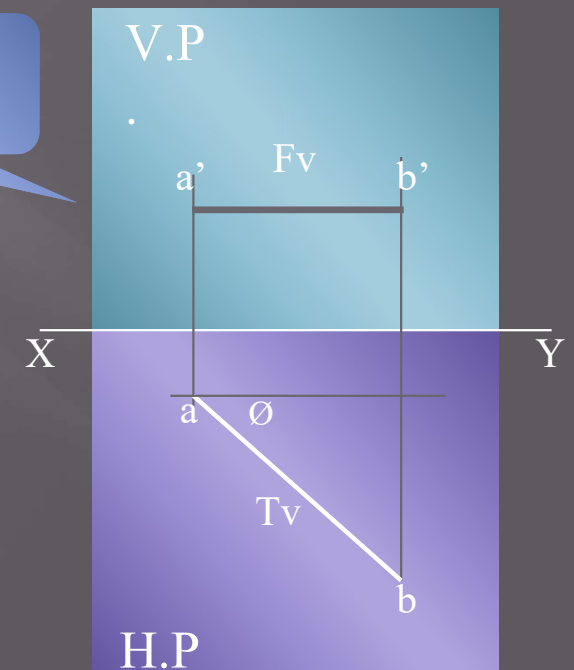
Orthographic Projections

4

**A Line inclined to Vp
and
parallel to Hp**

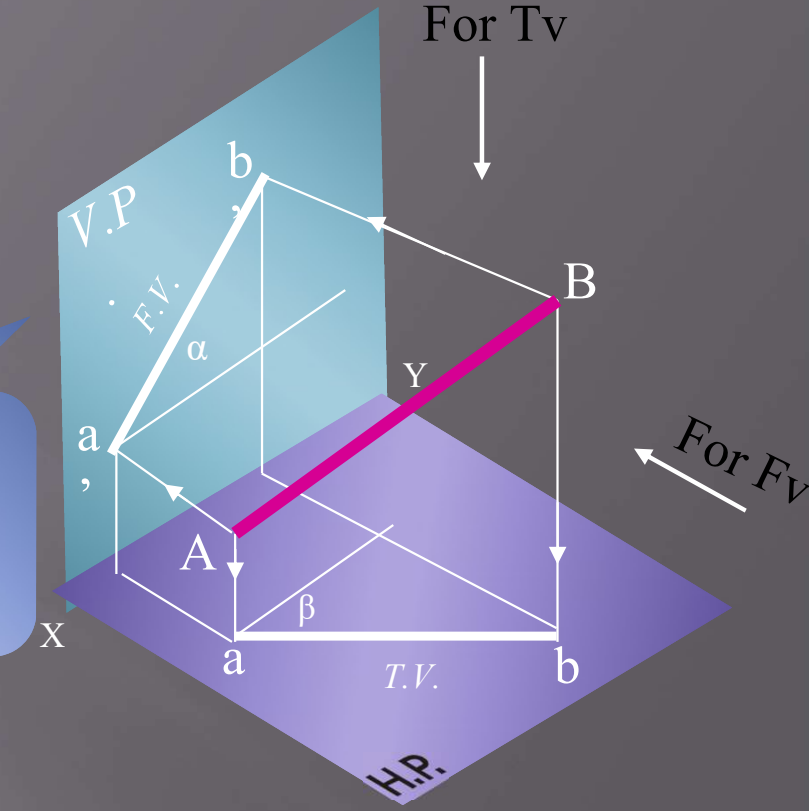
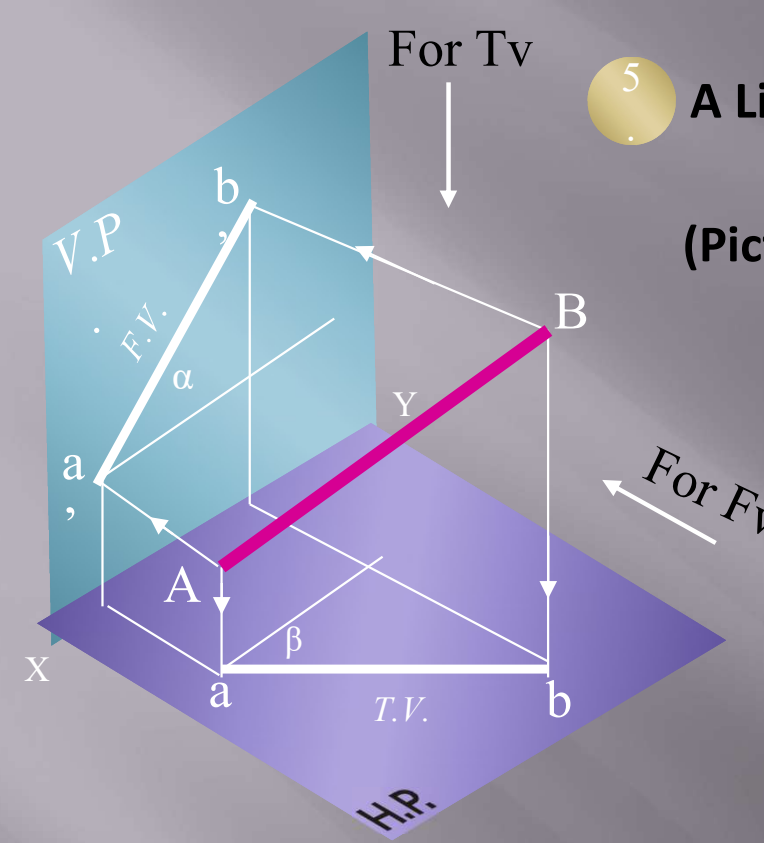


**TV inclined to XY
FV parallel to XY.**

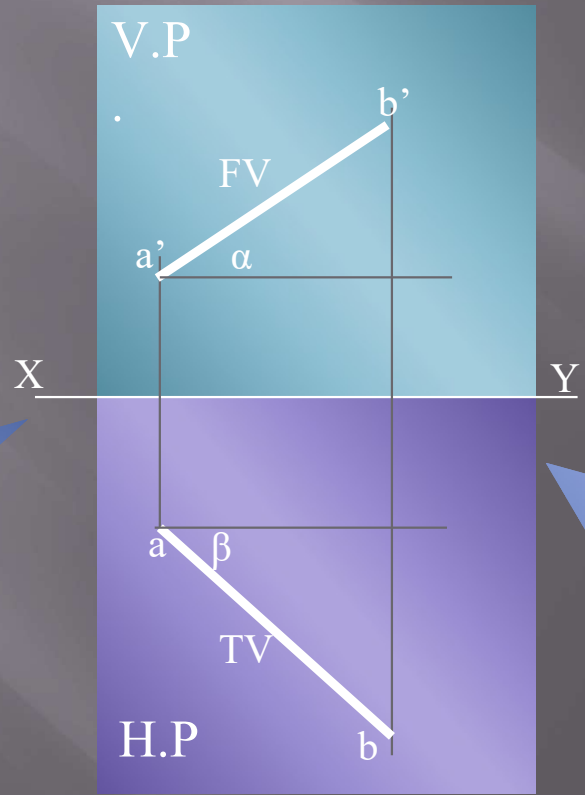


5

A Line inclined to both Hp and Vp (Pictorial presentation)



On removal of object
i.e. Line AB
Fv as a image on Vp.
Tv as a image on Hp,



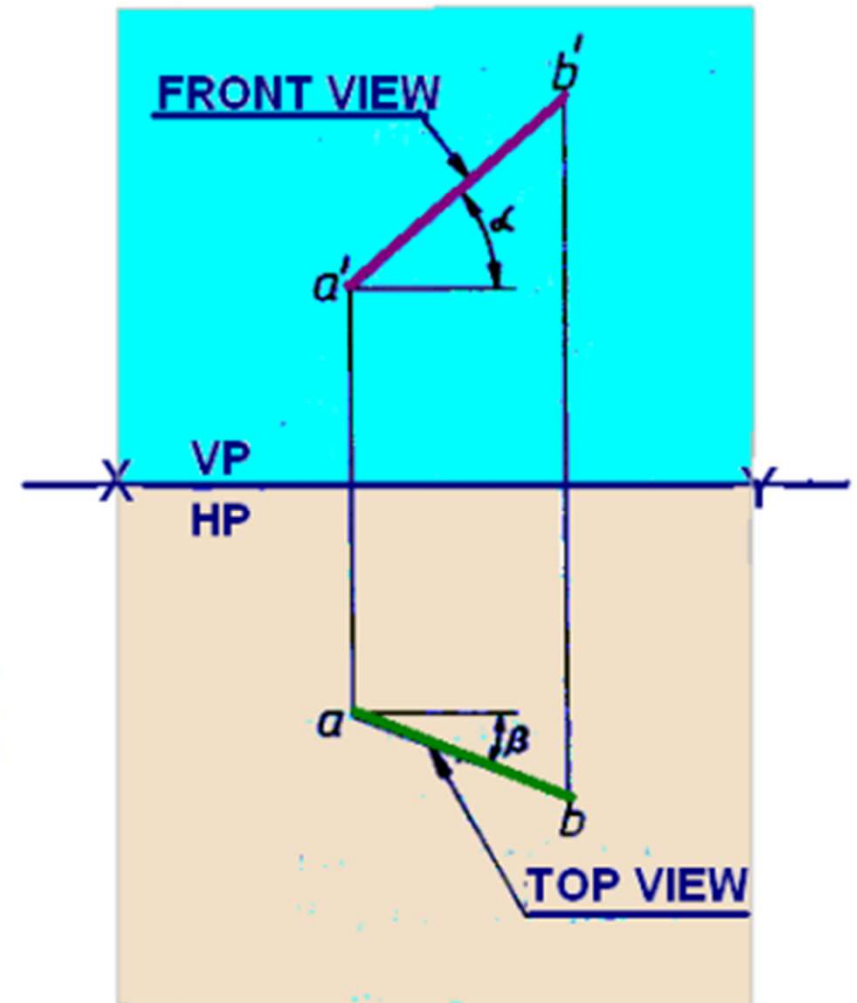
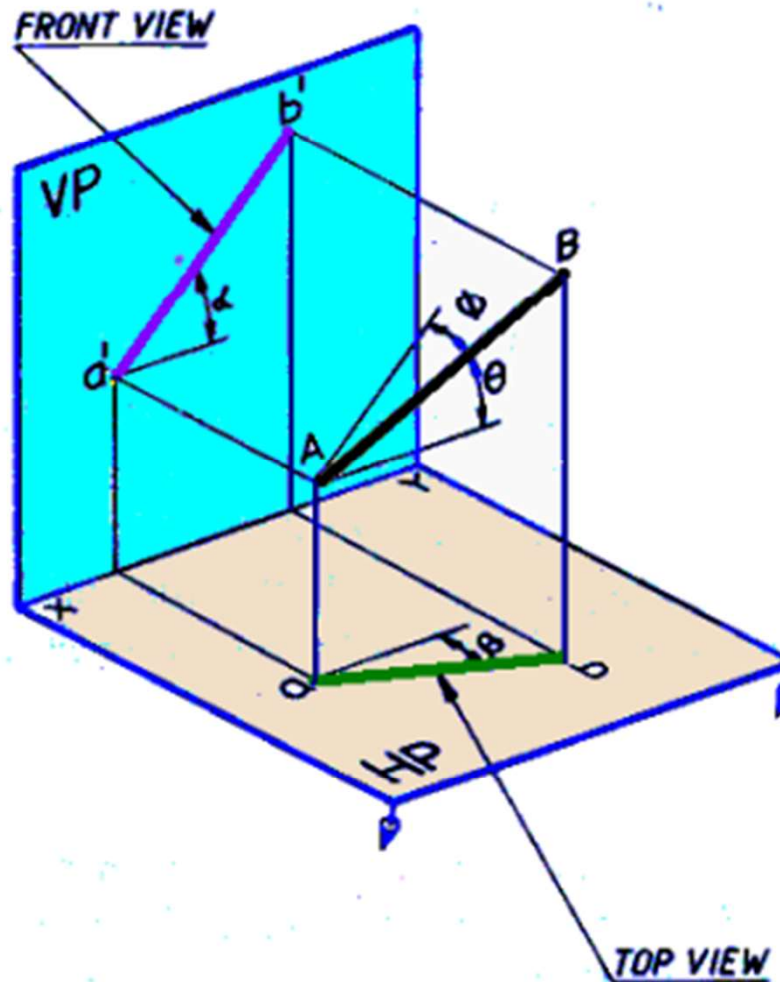
Orthographic Projections
Fv is seen on Vp clearly.
*To see Tv clearly, HP is
rotated 90° downwards,
Hence it comes below xy.*

Note These Facts:-
Both Fv & Tv are inclined to
xy.
(No view is parallel to xy)
Both Fv & Tv are reduced
lengths.
(No view shows True Length)

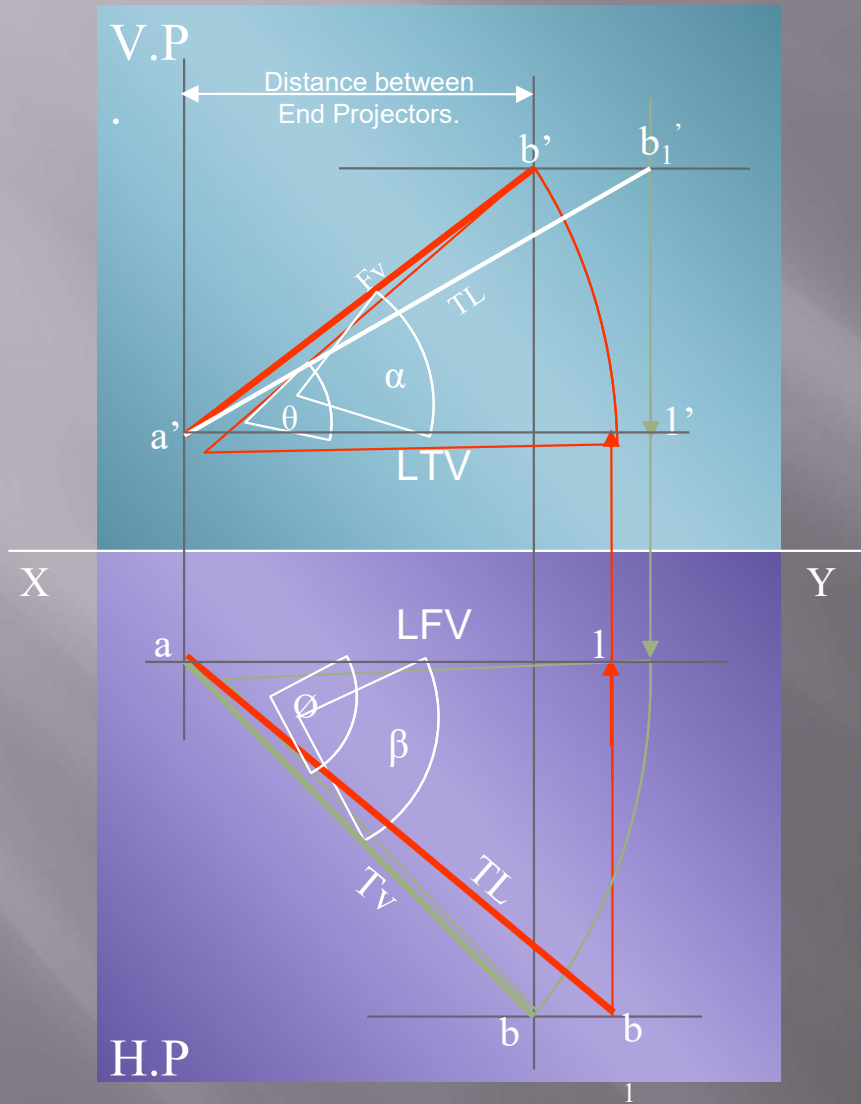
Line inclined to HP and VP

Apparent Inclinations: α and β

Apparent Lengths: ab , $a'b'$



The most important diagram showing graphical relations among all important parameters of this topic. Study and memorize it as a *CIRCUIT DIAGRAM* And use in solving various problems.



- 1) True Length (TL) – $a' b_1'$ & $a b$
- 2) Angle of TL with Hp – θ
- 3) Angle of TL with Vp – ϕ
- 4) Angle of FV with xy – α
- 5) Angle of TV with xy – β
- 6) LTV (length of TV) – Component
- 7) LFV (length of FV) – Component
- 8) Position of A- Distances of a & a' from xy
- 9) Position of B- Distances of b & b' from xy
- 10) Distance between End Projectors

Important
TEN parameters
to be remembered
with Notations
used here onward

NOTE this

θ & α Construct with a'
 $\emptyset$ & β Construct with a
 b' & b_1' on same locus.
 b & b_1 on same locus.

Also Remember

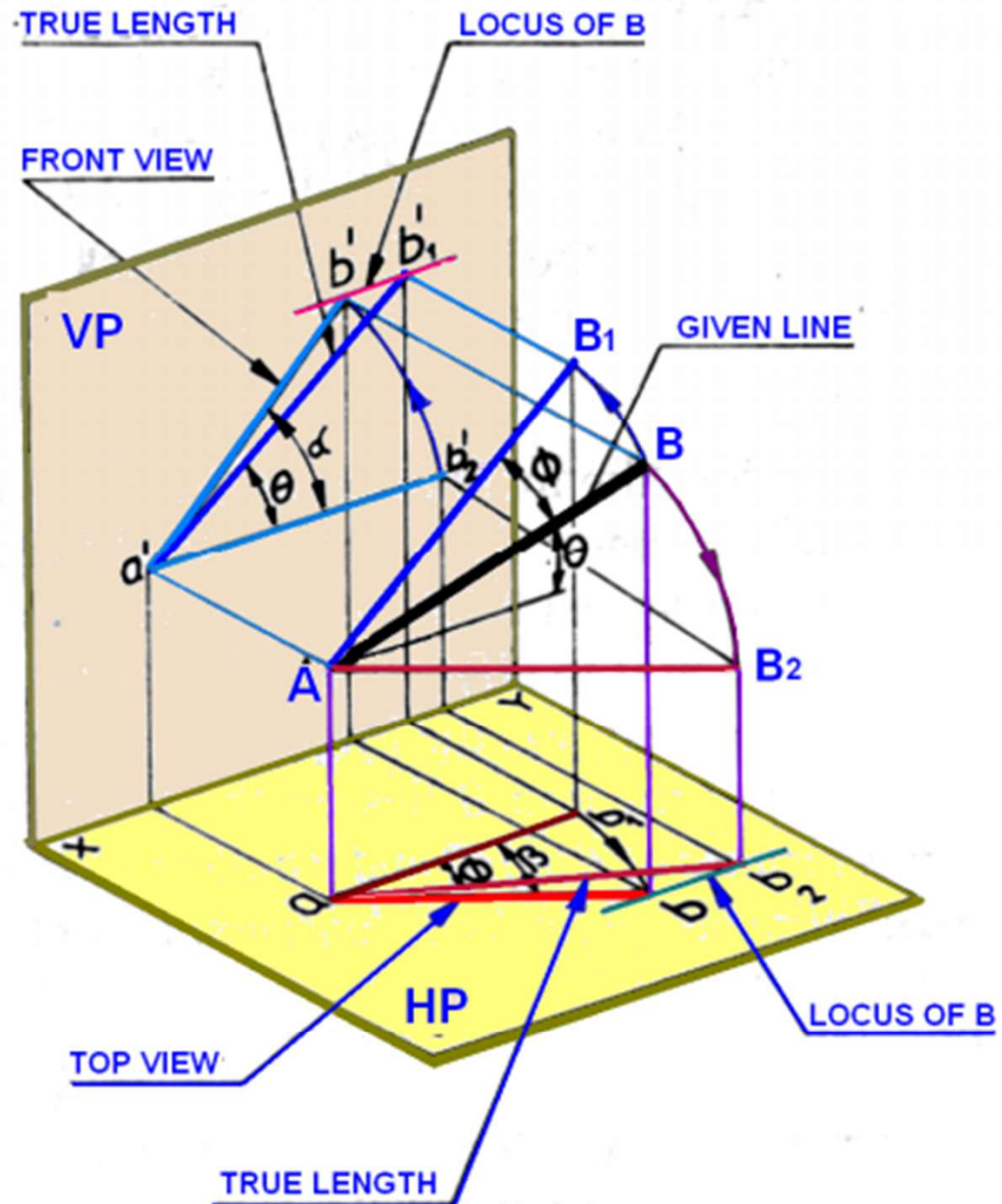
True Length is never rotated. It's horizontal component is drawn & it is further rotated to locate view.

Views are always rotated, made horizontal & further extended to locate TL, θ & $\emptyset$

Line inclined to HP and VP.....

Draw the projections of a line AB inclined to both HP and VP, whose true length and true inclinations and locations of one of the end points, say A are given.

Since the line AB is inclined at θ to HP and ϕ to VP – its top view ab and the front view a'b' are not in true lengths and they are also not inclined at angles θ to HP and ϕ to VP.



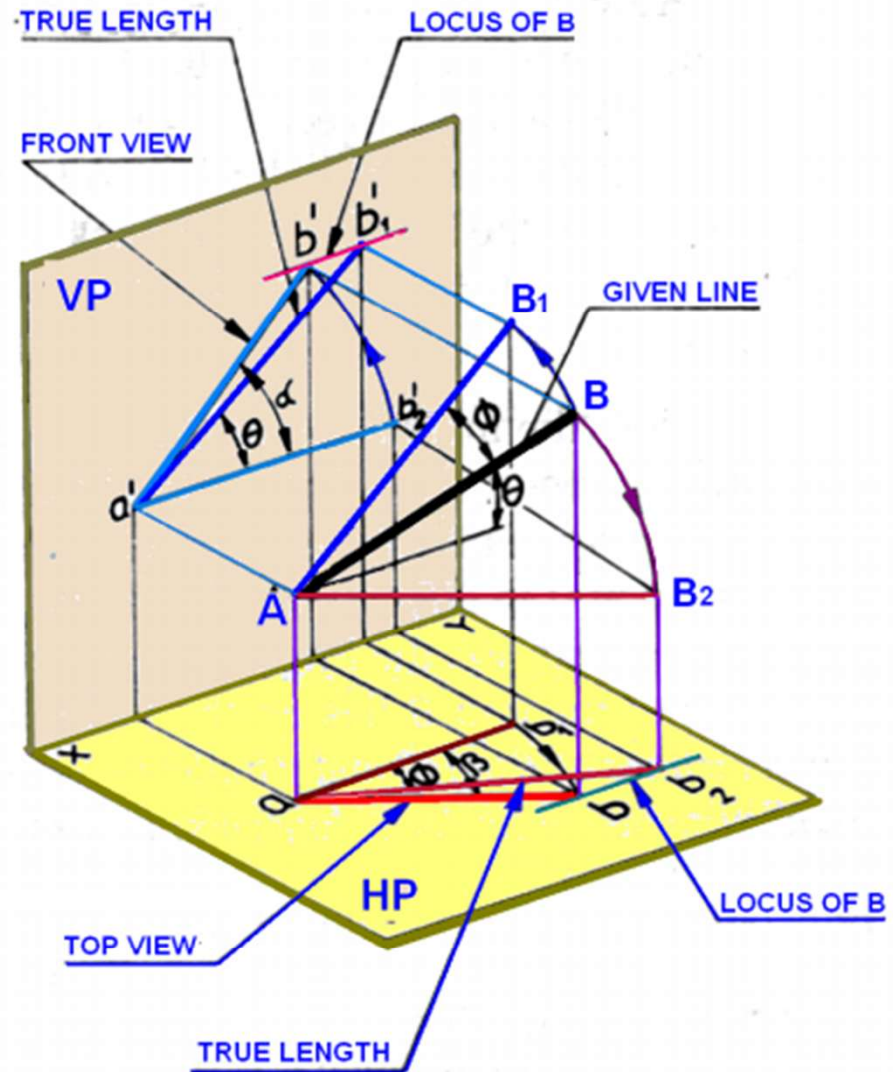
Line inclined to HP and VP.....

Step 1: Rotate the line AB to make it parallel to VP.

Rotate the line **AB** about the end **A**, keeping θ , the inclination of **AB** with **HP** constant till it becomes parallel to **VP**. This rotation of the line will bring the end **B** to the new position **B₁**.

AB₁ is the new position of the line **AB** when it is inclined at θ to **HP** and parallel to **VP**.

Project **AB₁** on **VP** and **HP**. Since **AB₁** is parallel to **VP**, **a'b'₁'**, the projection of **AB₁** on **VP** is in true length inclined at θ to the **XY** line, and **ab₁**, the projection of **AB₁** on **HP** is parallel to the **XY** line. Now the line is rotated back to its original position **AB**.



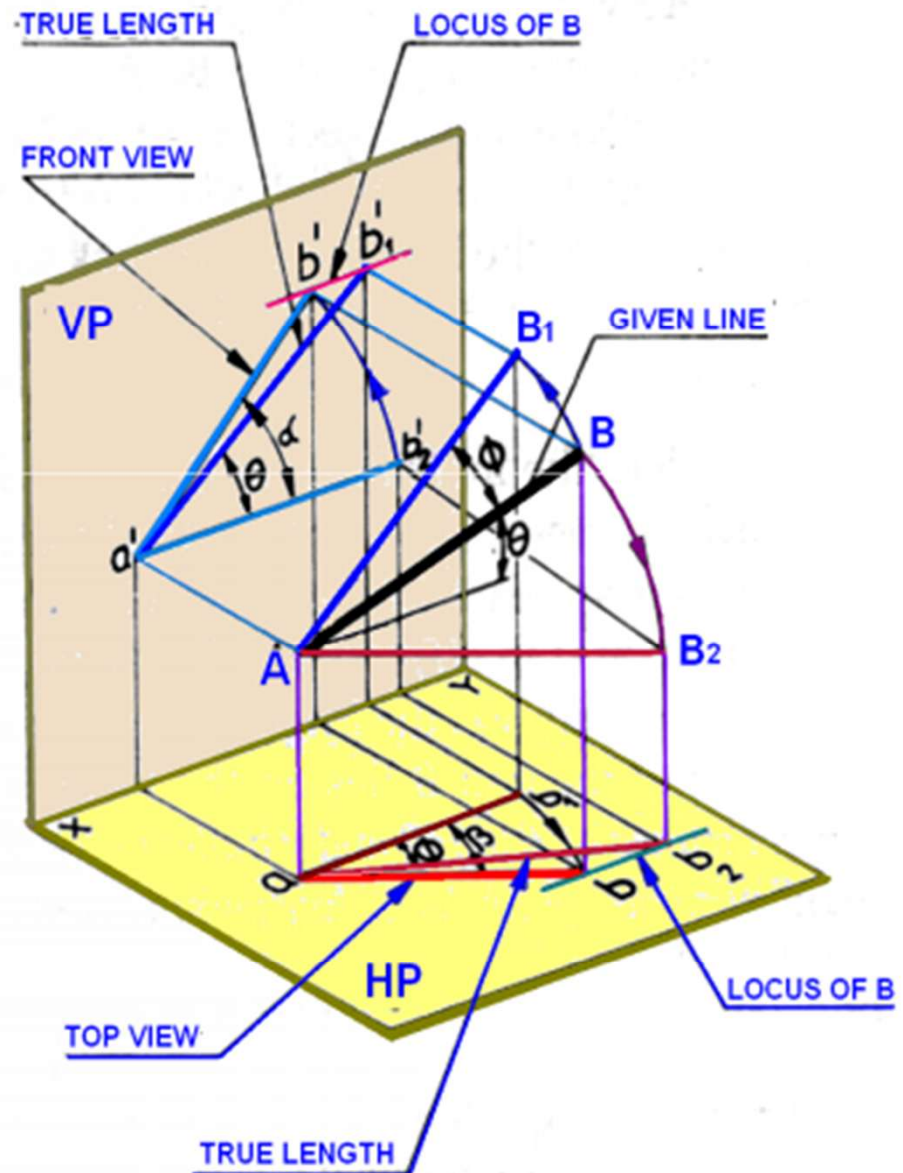
Line inclined to HP and VP.....

Step 2: Rotate the line AB to make it parallel to HP.

Rotate the line **AB** about the end **A** keeping ϕ the inclination of **AB** with **VP** constant, till it becomes parallel to **HP**. This rotation of the line will bring the end **B** to the second new Position **B₂**.

AB₂ is the new position of the line **AB**, when it is inclined at ϕ to **VP** and parallel to **HP**.

Project **AB₂** on **HP** and **VP**. Since **AB₂** is parallel to **HP**, **ab₂**, the projection of **AB₂** on **HP** is in true length inclined at ϕ to **XY** line, and **a'b₂'** the projection of **AB₂** on **VP** is parallel to **XY** line. Now the line is rotated back to its original position **AB**.



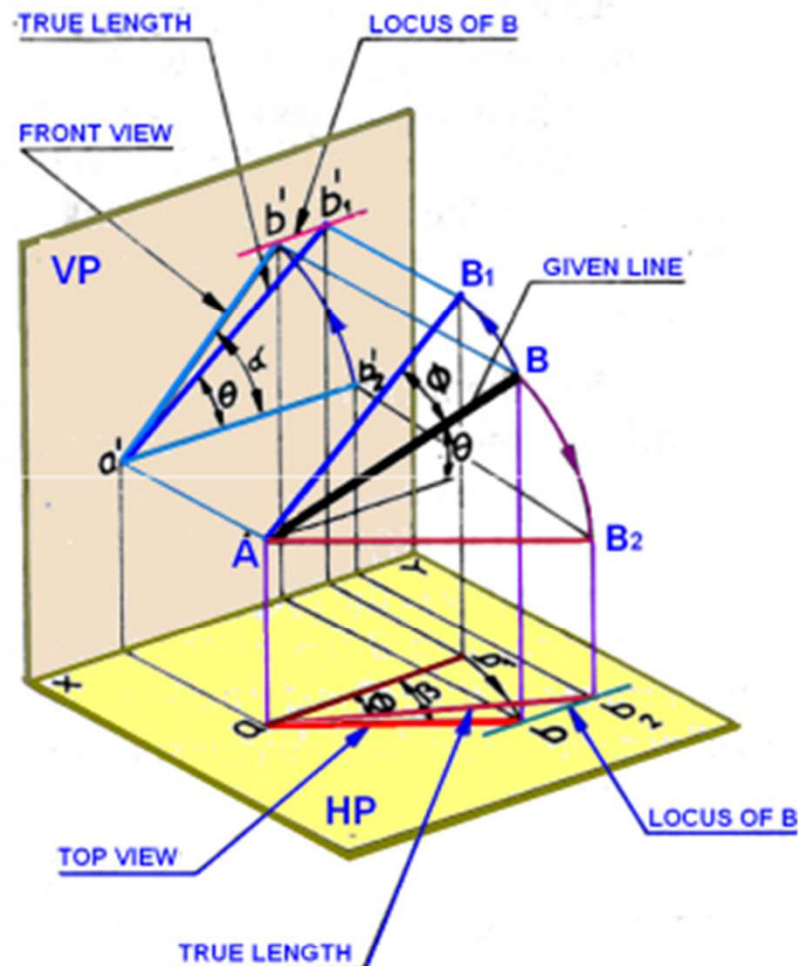
Line inclined to HP and VP.....

Step 3: Locus of end B in the front view

When the line AB is swept around about the end A keeping θ , the inclination of the line with the HP constant, by one complete rotation, the end B will always be at the same vertical height above HP, and the locus of the end B will be a circle which appears in the front view as a horizontal line passing through b' .

As long as the line is inclined at θ to HP, whatever may be the position of the line (i.e., whatever may be the inclination of the line with VP) the length of the top view will always be equal to ab_1 and in the front view the projection of the end B lies on the locus line passing through b_1' .

Thus ab_1 , the top view of the line when it is inclined at θ to HP and parallel to VP will be equal to ab and b' , the projection of the end B in the front view will lie on the locus line



GROUP (A)

GENERAL CASES OF THE LINE INCLINED TO BOTH HP & VP (based on 10 parameters).

PROBLEM 1)

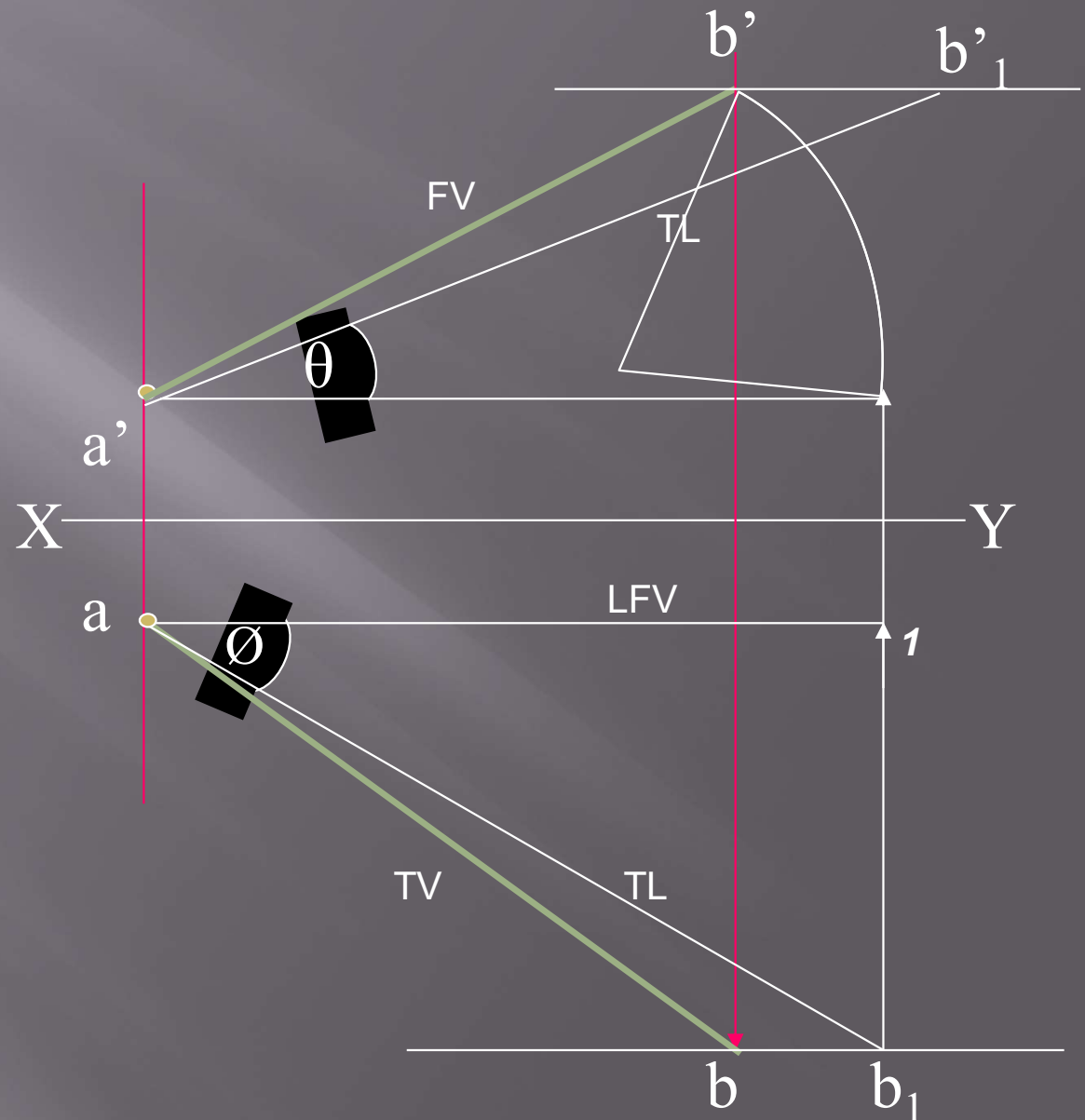
Line AB is 75 mm long and it is 30° & 40° Inclined to Hp & Vp respectively.

End A is 12mm above Hp and 10 mm in front of Vp.

Draw projections. Line is in 1st quadrant.

SOLUTION STEPS:

- 1) Draw xy line and one projector.
- 2) Locate a' 12mm above xy line & a 10mm below xy line.
- 3) Take 30° angle from a' & 40° from a and mark TL i.e. 75mm on both lines. Name those points b_1' and b_1 respectively.
- 4) Join both points with a' and a resp.
- 5) Draw horizontal lines (Locus) from both points.
- 6) Draw horizontal component of TL a b_1 from point b_1 and name it 1. (the length $a-1$ gives length of Fv as we have seen already.)
- 7) Extend it up to locus of a' and rotating a' as center locate b' as shown. Join a' b' as Fv.
- 8) From b' drop a projector down ward & get point b . Join a & b i.e. Tv.

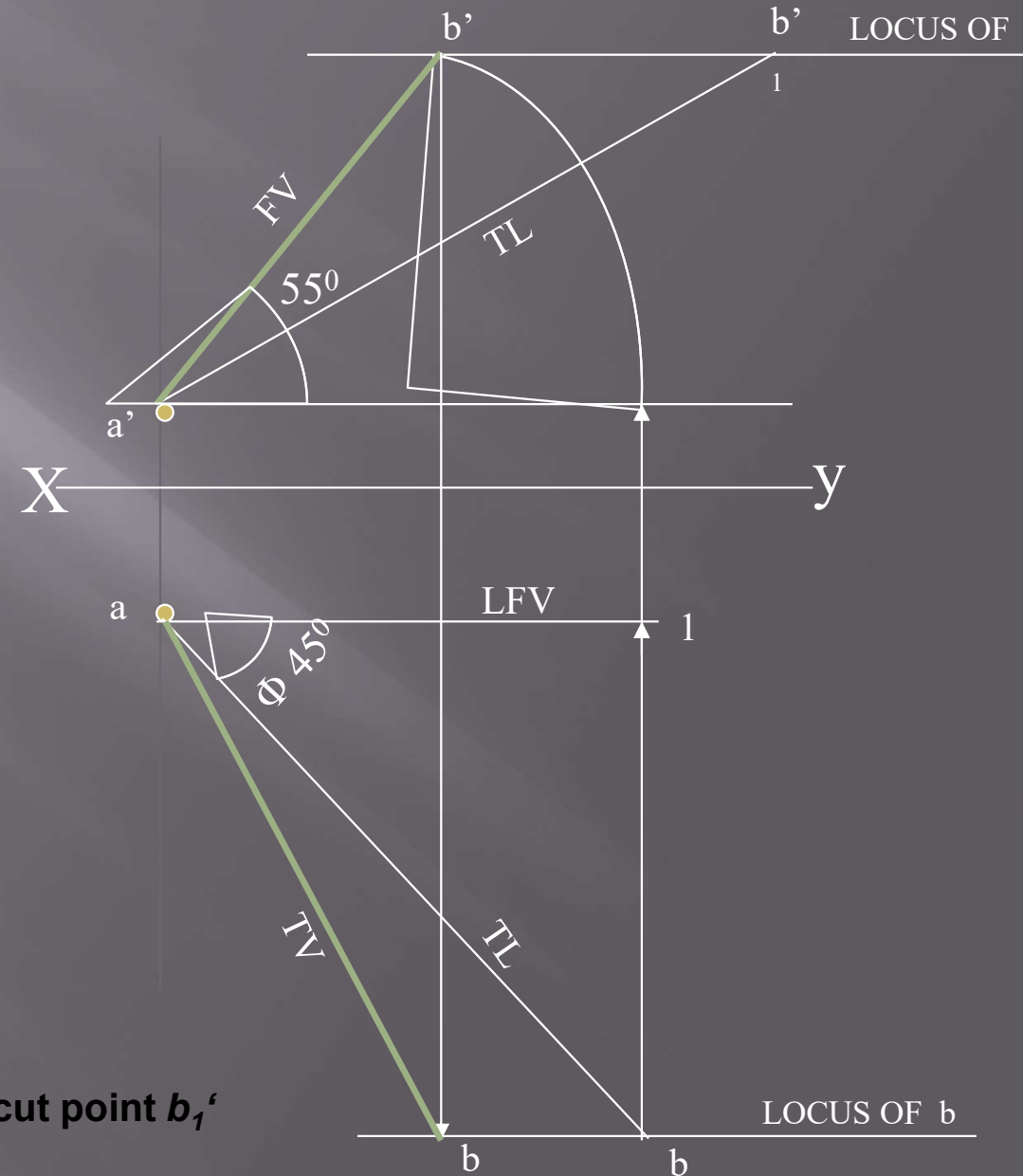


PROBLEM 2:

Line AB 75mm long makes 45° inclination with Vp while its Fv makes 55° . End A is 10 mm above Hp and 15 mm in front of Vp. If line is in 1st quadrant draw its projections and find its inclination with Hp.

Solution Steps:-

1. Draw x-y line.
2. Draw one projector for a' & a
3. Locate a' 10mm above x-y & a 15 mm below xy.
4. Draw a line 45° inclined to xy from point a and cut TL 75 mm on it and name that point b_1 . Draw locus from point b_1
5. Take 55° angle from a' for Fv above xy line.
6. Draw a vertical line from b_1 up to locus of a' and name it 1. It is horizontal component of TL & is LFV.
7. Continue it to locus of a' and rotate upward up to the line of Fv and name it b' . This $a'b'$ line is Fv.
8. Drop a projector from b' on locus from point b_1 and name intersecting point b . Line ab is Tv of line AB.
9. Draw locus from b' and from a' with TL distance cut point b_1'
10. Join $a'b_1'$ as TL and measure its angle at a' . It will be true angle of line with HP.

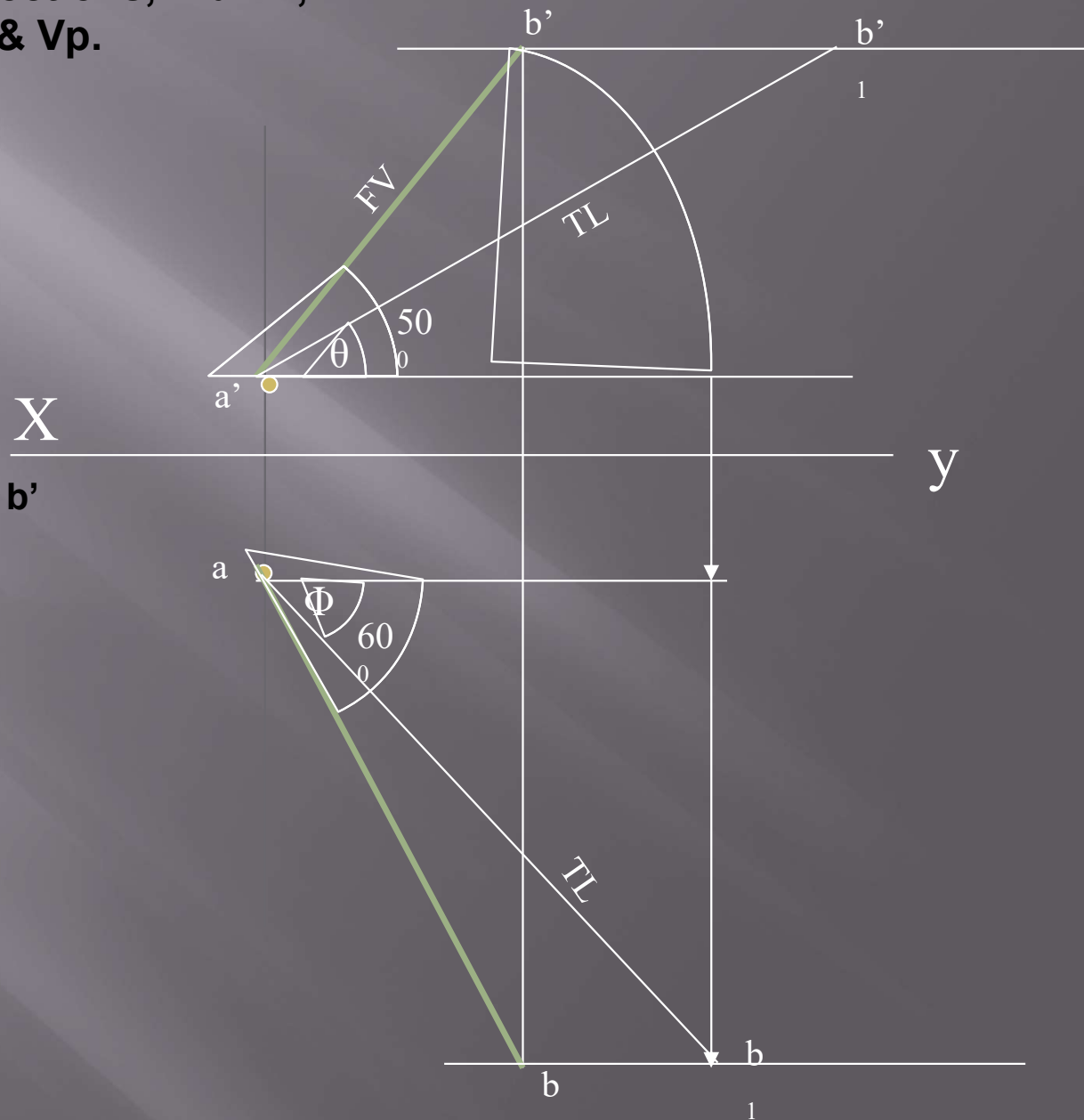


PROBLEM 3:

Fv of line AB is 50° inclined to xy and measures 55 mm long while it's Tv is 60° inclined to xy line. If end A is 10 mm above Hp and 15 mm in front of Vp, draw it's projections, find TL, inclinations of line with Hp & Vp.

SOLUTION STEPS:

1. Draw xy line and one projector.
2. Locate a' 10 mm above xy and a 15 mm below xy line.
3. Draw locus from these points.
4. Draw Fv 50° to xy from a' and mark b' Cutting 55mm on it.
5. Similarly draw Tv 60° to xy from a & drawing projector from b' Locate point b and join a b.
6. Then rotating views as shown, locate True Lengths ab_1 & $a'b_1'$ and their angles with Hp and Vp.



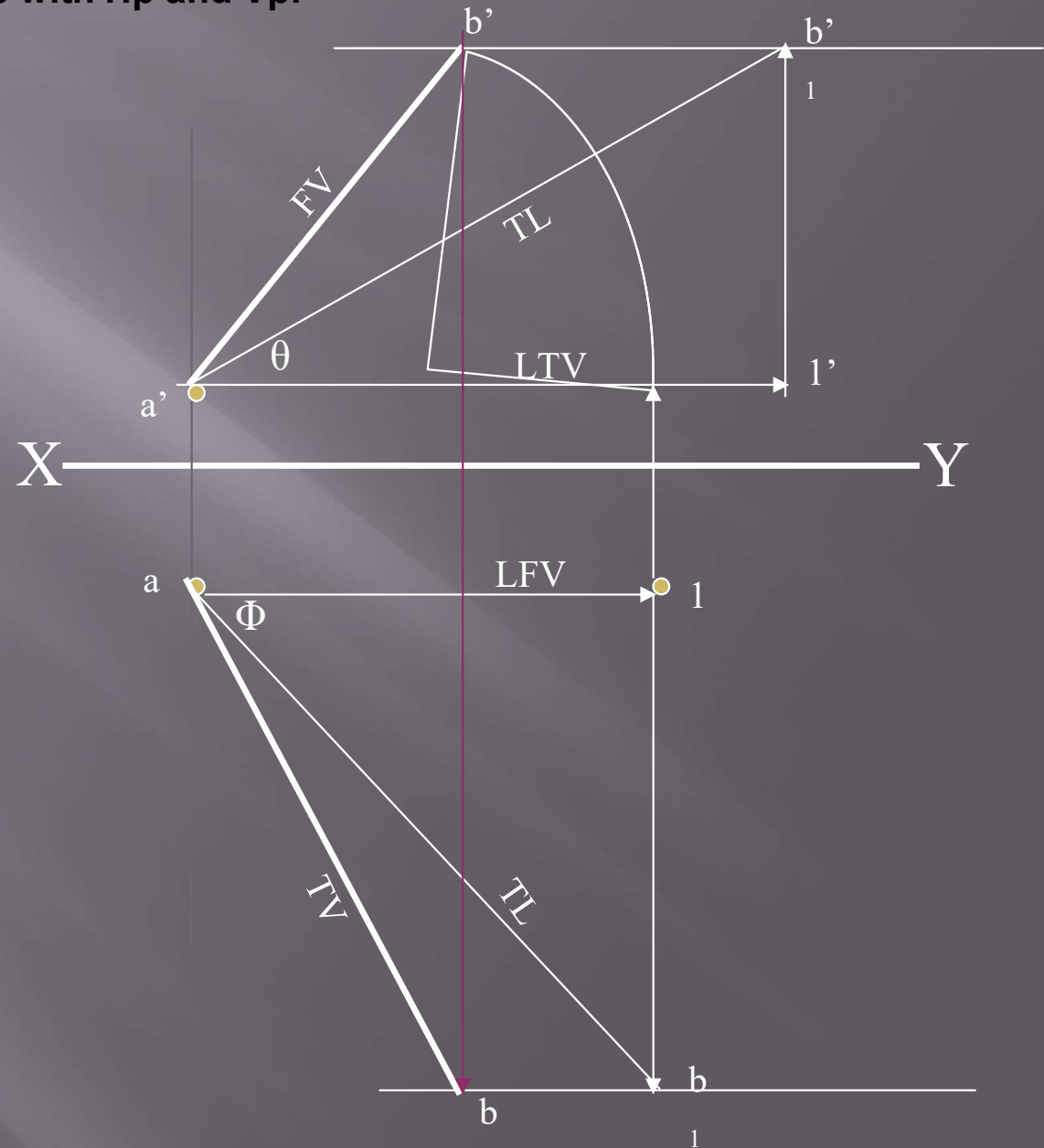
PROBLEM 4 :-

Line AB is 75 mm long .It's Fv and Tv measure 50 mm & 60 mm long respectively.

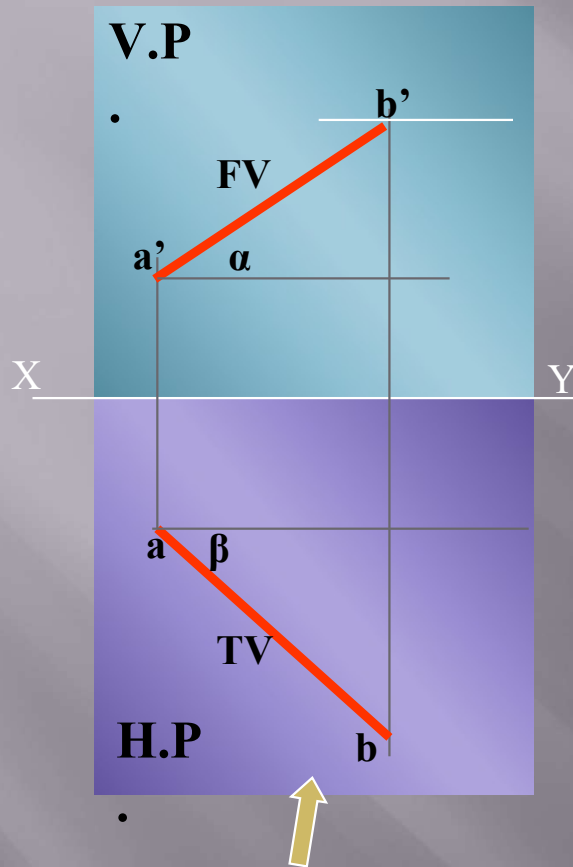
End A is 10 mm above Hp and 15 mm in front of Vp. Draw projections of line AB if end B is in first quadrant. Find angle with Hp and Vp.

SOLUTION STEPS:

1. Draw xy line and one projector.
2. Locate a' 10 mm above xy and a 15 mm below xy line.
3. Draw locus from these points.
4. Cut 60mm distance on locus of a' & mark $1'$ on it as it is LTV.
5. Similarly cut 50mm on locus of a and mark point 1 as it is LFV.
6. From $1'$ draw a vertical line upward and from a' taking TL (75mm) in compass, mark b'_1 point on it. Join a' b'_1 points.
7. Draw locus from b'_1
8. With same steps below get b_1 point and draw also locus from it.
9. Now rotating one of the components i.e. a-1 locate b' and join a' with it to get Fv.
10. Locate tv similarly and measure Angles θ & Φ

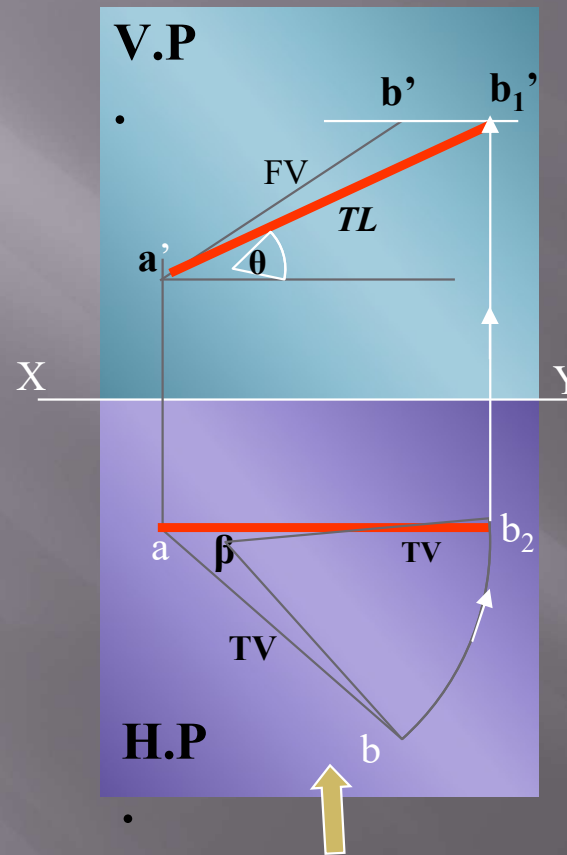


Orthographic Projections
Means Fv & Tv of Line AB
are shown below,
with their apparent Inclinations
 α & β



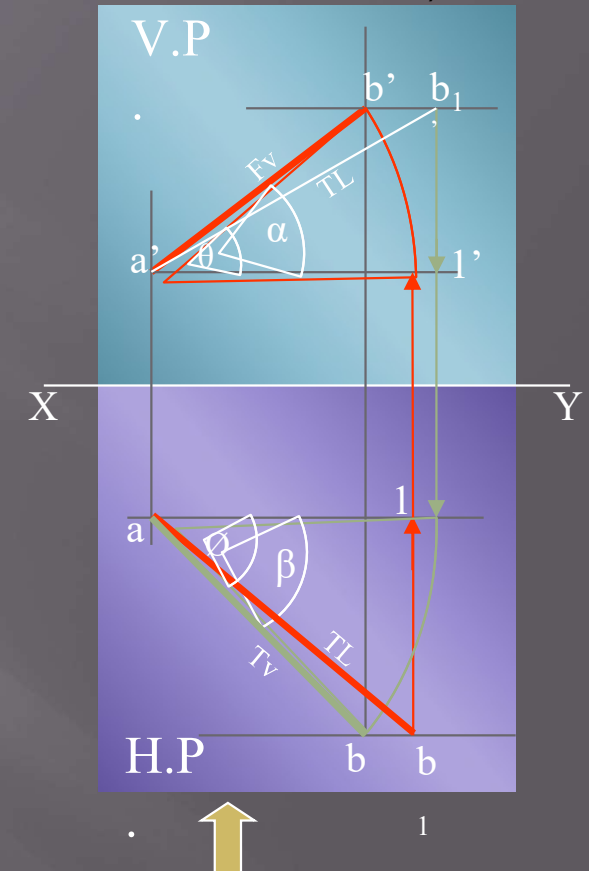
*Here TV (ab) is not // to XY line
Hence it's corresponding FV
 $a' b'$ is not showing
True Length &
True Inclination with Hp.*

Note the procedure
When Fv & Tv known,
How to find True Length.
(Views are rotated to determine
True Length & its inclinations
with Hp & Vp).



*In this sketch, TV is rotated
and made // to XY line.
Hence it's corresponding
FV $a' b_1'$ is showing
True Length
&
True Inclination with Hp.*

Note the procedure
When True Length is known,
How to locate Fv & Tv.
(Component a-1 of TL is drawn
which is further rotated
to determine Fv)



*Here a-1 is component
of TL $a b_1$ gives length of Fv.
Hence it is brought Up to
Locus of a' and further rotated
to get point b' . $a' b'$ will be Fv.
Similarly drawing component
of other TL ($a' b_1'$) Tv can be drawn.*

GENERAL CASES OF THE LINE INCLINED TO BOTH HP & VP
(based on 10 parameters).

PROBLEM

Line AB is 75 mm long and it is 30° & 40° Inclined to Hp & Vp respectively.
End A is 12mm above Hp and 10 mm in front of Vp.

Draw projections. Line is in 1st quadrant.

SOLUTION STEPS:

- 1) Draw xy line and one projector.
- 2) Locate a' 12mm above xy line & a 10mm below xy line.
- 3) Take 30° angle from a' & 40° from a and mark TL i.e. 75mm on both lines. Name those points b_1' and b_1 respectively.
- 4) Join both points with a' and a resp.
- 5) Draw horizontal lines (Locus) from both points.
- 6) Draw horizontal component of TL a b_1 from point b_1 and name it 1.
(the length $a-1$ gives length of Fv as we have seen already.)
- 7) Extend it up to locus of a' and rotating a' as center locate b' as shown.
Join $a' b'$ as Fv.
- 8) From b' drop a projector down ward & get point b . Join a & b i.e. Tv.

